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REGULAR PAPER

SentiVol-GA: a volatility-scaled genetic fusion of predictive models and financial sentiment for adaptive stock forecasting

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Abstract
SentiVol-GA is an evolvable hybrid framework for stock price forecasting. This model integrates statistical models, deep learning architectures, financial sentiment analysis, and volatility-aware optimization through a Genetic Algorithm (GA). This framework combines five predictive models—Linear Regression, LSTM, GRU, Bi-LSTM, and ARIMA—with sentiment signals extracted from FinBERT, VADER, and the Loughran–McDonald dictionary. Its core innovation lies in a volatility-scaling mechanism that adjusts sentiment impact during turbulent periods, while the GA periodically optimizes model and sentiment weights to minimize forecasting error thus sustaining the model's predictive stability. This empirical evaluation across eight Indian IT-sector stocks—covering large, mid- and small-cap segments, demonstrates up to 12% improvement in R^2, 30–60% reduction in RMSE, and 20–35% gain in tolerance-based accuracy over classical and deep-learning baselines. Nonparametric Friedman and Wilcoxon tests ($p < 0.05$) confirm that these performance gains are statistically significant. SentiVol-GA offers a volatility-aware, sentiment-driven, and dynamically optimized approach that is scalable, interpretable, and robust for intelligent financial decision-making.

Keywords SentiVol-GA · Adaptive stock forecasting · Sentiment analysis · Volatility scaling · Genetic algorithm · Deep learning · Financial time series

1 Introduction

The prediction of stock prices is a challenging task due to the inherently volatile, nonlinear, and multifactorial nature of financial markets. Stock movements are influenced not only by historical patterns and technical indicators but also by rapidly evolving factors such as investor sentiment, global news, and macroeconomic shifts. Traditional methods like technical charting and linear statistical forecasting have been widely used, but they often fall short in adapting to real-time, high-dimensional data streams and in capturing the complex dependencies inherent in financial time series data [1].

In recent years, machine learning (ML) and deep learning (DL) techniques have emerged as powerful tools for financial forecasting, owing to their capacity to model both linear and nonlinear patterns from large volumes of data [2, 3]. These models, especially recurrent networks like LSTM and GRU, are capable of capturing long-term temporal dependencies, making them suitable for financial time series forecasting [4]. However, most ML and DL-based methods focus solely on historical price and volume data, thereby neglecting qualitative factors such as sentiment or market mood, which have been proven to significantly impact short-term price fluctuations.

To address this gap, we propose SentiVol-GA, a volatility-enhanced hybrid stock prediction framework that combines deep learning, statistical modeling, financial sentiment analysis, and Genetic Algorithm (GA)-based optimization. This approach leverages a modular ensemble of five models—Linear Regression, LSTM, GRU, Bi-LSTM, and ARIMA—to capture diverse forecasting perspectives. Each model is trained to forecast the next-day closing price, and their

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predictions are combined using a Genetic Algorithm that adaptively optimizes the model weights over time.

In parallel, we integrate sentiment analysis using three complementary tools: FinBERT, a transformer-based model fine-tuned for financial text [5]; VADER, a rule-based sentiment analyzer suitable for short texts like headlines and tweets [6]; and the Loughran–McDonald financial dictionary, which provides domain-specific word classification to reduce misclassification in financial documents [7]. These sentiment scores are aggregated and scaled using a volatility term derived from recent price fluctuations, ensuring that sentiment influence is modulated under turbulent market conditions.

Recent advancements in deep learning have significantly improved the predictive modeling of financial markets, particularly in stock price forecasting. Integrated deep learning frameworks have shown strong gains by fusing multiple market signals within a unified predictive architecture [8]. Deep learning models continue to demonstrate strong predictive advantages in financial time series, with LSTM architectures effectively capturing long-horizon dependencies and outperforming traditional statistical baselines in return forecasting tasks [9]. Complementary advances in nonlinear representation learning further show that combining deep feature extraction with sequential modeling can significantly enhance forecasting stability in complex market environments [10], while multimodal deep learning models that fuse price data with textual news have been shown to significantly improve stock-market prediction accuracy [11]. Beyond forecasting, deep learning has also been employed for asset pricing and feature extraction in complex market environments [12, 13]. Emerging domains like crypto currency forecasting have benefited from similar architectures, demonstrating versatility across financial assets [14]. Deep learning–based integrated frameworks have recently shown strong gains in stock-movement prediction by merging multiple market signals within a single architecture [15]. Complementary hybrid models—such as the Navier–Stokes-based iForest-Boruta SHAP-Prophet system—further demonstrate improved predictive stability in complex Indian market conditions [16]. Context-aware attention architectures have shown strong effectiveness in handling noisy, fast-evolving textual streams by capturing subtle linguistic cues in complex information environments [17]. Complementary evidence further indicates that decomposition-based predictive models can successfully characterize macroeconomic shocks, uncertainty regimes, and speculative bubble dynamics in Indian equity markets, offering deeper structural insights into market behavior [18]. Hybrid models incorporating attention mechanisms and probabilistic learning further enhance prediction accuracy by aligning textual sentiment with technical indicators [19, 20]. The advantages of integrating natural language processing with quantitative models have also been highlighted by recent surveys that have compiled research on sentiment-based forecasting techniques [21]. The goal of the Chen et al. study [22] was to use anticipatory computing to investigate public sentiment and emotion from news reports. Following that, it makes predictions about the direction of the stock market, which can serve as a model for similar industries.

Our key innovation lies in this volatility-scaled dynamic weighting mechanism, powered by GA, which differentiates the proposed hybrid from fixed-weight ensembles or static predictors. This makes the system adaptive not only to structural market shifts but also to short-term sentiment dynamics. Monthly re-optimization of weights enables the framework to stay aligned with recent trends, ensuring sustained accuracy and responsiveness.

Experiments conducted on eight Indian IT-sector stocks—spanning large-cap (TCS, Infosys, HCL Tech, Wipro), mid-cap (Persistent Systems, Sasken Technologies), and small-cap companies (Policy Bazar, Quick Heal)—demonstrate that the SentiVol-GA model significantly outperforms standalone models in terms of R^2, RMSE, MAE, and tolerance-based accuracy. Statistical validation via Friedman and Wilcoxon tests further confirms the robustness and superiority of the hybrid approach over traditional techniques. By effectively combining quantitative modeling with sentiment analysis and intelligent optimization, SentiVol-GA offers a scalable and real-world–ready solution for adaptive stock price forecasting.

2 Related works

We have organized the body of existing research into two primary streams to better comprehend the current state of stock price forecasting: one focusing on sentiment analysis in financial forecasting and the other on sentiment-free deep learning models. This classification shows how each area has grown, yet highlights the need to blend number-based signals with mood-sensitive inputs for stronger results. To establish a strong foundation for the proposed hybrid framework, we further present a Comparative Summary of Reviewed Literature, which encapsulates key studies along with their methodologies, techniques employed; outcomes achieved, and recognized limitations.

2.1 Studies related to stock price prediction using machine learning models

Recent advancements in deep learning have significantly influenced the domain of stock market prediction by capturing both linear and complex nonlinear dependencies in time series data. Patra and Mohanty [23] proposed a hybrid LSTM–GRU framework tailored for the S&P 500 index.

Their model demonstrated superior predictive accuracy and reduced forecasting error compared to standalone LSTM or GRU models. Shi et al. [24] advanced this by integrating CNN–LSTM with XGBoost in a hybrid model that leverages spatial, temporal, and feature-level representations. Their ensemble achieved better generalization and reliability under heterogeneous market conditions. Thakkar and Chaudhari [25] had done survey to check applicability of genetic algorithms for stock market prediction. They found that Genetic algorithms help boost stock prediction accuracy by tuning parameters and selecting key features. When combined with models like LSTM or TCN, they perform better than standalone models. However, issues like high computational cost and over fitting still need attention. Similarly, Barua et al. [26] evaluated deep learning architectures such as RNN, GRU, LSTM, CNN, and Attention-LSTM across Indian stocks like TCS, ICICI, and Reliance. Their results showed that GRU and CNN achieved higher accuracy across most datasets, while Attention-LSTM performed well in high-volatility contexts. Awad et al. [27] further enhanced performance by exploring ensemble deep learning models like CNN–LSTM and GRU–CNN, which showed resilience and generalization capability during unstable market periods like the COVID-19 crisis. Shankar et al. compared ARIMA, LSTM, and UCM models for predicting Tata Motors stock prices. UCM gave the best results, followed by LSTM. ARIMA performed the worst. The study was limited to one company and did not include external factors like news or sentiment [28].

Although these works validate the strength of deep and hybrid learning, they typically rely on fixed weighting schemes and lack dynamic adaptability to market sentiment and volatility. These gaps underline the motivation for our study, which proposes a volatility-aware hybrid framework with GA-driven weight optimization for adaptive stock price prediction.

2.2 Studies related to market sentiment analysis in financial forecasting

Investor sentiment, particularly extracted from financial news and social media platforms, has emerged as a critical factor in predicting stock market movements. Recent advances in NLP and the advent of transformer models have revolutionized sentiment analysis in finance.

Padmanayana et al. [29] studied how Twitter sentiment can help in predicting stock prices more accurately. They have used a Naïve Bayes classifier to classify tweets, and then, this sentiment data was combined with past stock prices and given to an XGBoost model for prediction. The model gave good results with 89.8% accuracy. Jiang et al. [30] proposed a FinBERT-enhanced prediction model, to extract sentiment from financial news headlines, and fed this into an LSTM network to forecast stock movement. Compared against standard BERT, pure LSTM, and classical ARIMA, they found FinBERT + LSTM delivered the best predictive performance, outperforming both sentiment-free LSTM and ARIMA by a good margin. Dhabe et al. [31] built a web application that gives stock trend prediction by combining LSTM-based price forecasting with Twitter sentiment data. The addition of Twitter-derived features—like percentages of positive and negative tweets, gives a better opinion on a particular stock for the users. In their paper, Varghese and Mohan [32] built a four-stage analytical model: First, they have web-scraped financial news articles and then sentiment score is calculated via a modified VADER algorithm (fine-tuned for financial context ranging from −1 to +1, where −1 is very negative and +1 means very positive),then they applied both parametric (Granger causality) and nonparametric (Shannon & Renyi's transfer entropy) causation tests. This allowed them to identify the direction and strength of the information flow between news sentiment and stock price movements.

Chauhan et al. [1] integrated financial headline sentiment with LSTM and reported improvements in accuracy, although they did not dynamically adjust sentiment influence over time. Wang et al. [2] used multi-source inputs including sentiment, yet applied static weights, limiting model flexibility in real-time decision-making.

These studies collectively show that while sentiment enhances prediction, existing systems often fail to adaptively scale its impact based on market volatility or recent events. This reinforces the novelty and necessity of our proposed approach, which uniquely combines real-time sentiment and volatility-adjusted forecasting with dynamic GA-based weight optimization.

2.3 Comparative summary of reviewed literature

We reviewed 21 studies to track the development of machine learning, sentiment analysis, and optimization based stock forecasting methods. Table 1 shows the variety of models, from LSTM and CNN predictors to sentiment-enhanced and GA optimized systems. Although these approaches brought progress, many missed combining volatility patterns, diverse sentiment data, while coordinating multiple models effectively. Our comparative analysis reveals this shortfall, highlighting the novelty of our Hybrid Model which combines statistical, deep learning, and sentiment features in a volatility-scaled, GA-optimized framework for real-time financial forecasting.

The above comparative overview highlights key advancements in combining machine learning, deep learning, and sentiment analysis for stock price forecasting. It also exposes a persistent limitation in most prior studies: the lack of dynamic adaptation to market sentiment and volatility. Our work addresses these gaps by introducing SentiVol-GA, a

Table 1 Comparative Literature Review on Stock Price Prediction and Sentiment Analysis

Author(s)	Title	Methodology used	Techniques used	Results	Limitations
Chauhan et al. [1]	A novel deep learning model for stock market prediction using a sentiment analysis system…	Proposed SenT-In, a sentiment-aware deep learning model combining market data with news sentiment	CNN and GRU networks compute a daily sentiment index; an attention mechanism fuses the sentiment with price time series	Outperformed baseline models (GRU, LSTM, CNN, SVM): ~ + 9% accuracy, + 7% F1, + 13% AUC-ROC on average	Tested on select indices; adds model complexity
Wang et al. [2]	A hybrid attention-based LSTM-CNN model for stock price forecasting…	Developed a hybrid DL model using heterogeneous data (e.g. prices, news) with attention	Stack of LSTM and CNN layers with an attention mechanism to highlight important features from multi-source inputs	Reported improved accuracy over single-model approaches	Complex architecture; requires careful tuning
Sezer et al. [3]	Financial time series forecasting with deep learning: a systematic literature review (2005–2019)	Conducted a systematic literature review of DL methods in finance	Categorized studies by forecasting target (indices, forex, etc.) and DL model type (CNN, DBN, LSTM, etc.)	Found deep learning often outperforms traditional ML in financial forecasting; summarized trends and research gaps	Review covers up to 2019; no new experiments
Hiransha et al. [4]	NSE stock market prediction using deep-learning models	Empirical comparison of four neural network models on stock data (NSE/NYSE)	Trained MLP, vanilla RNN, LSTM and CNN on historical prices	CNN model gave the lowest error, outperforming RNN/LSTM and linear models	Short-term prediction only; no external sentiment or news data
Araci [5]	FinBERT: financial sentiment analysis with pre-trained language models	Developed FinBERT, a BERT-based model fine-tuned on financial text for sentiment analysis	Transformer (BERT) pre-trained on generic data, then fine-tuned on financial domain corpora	Outperforms prior state-of-art on financial sentiment benchmarks; higher precision/F1 even with limited training data	Focus is sentiment labeling, not stock forecasting
Hutto & Gilbert [6]	VADER: a parsimonious rule-based model for sentiment analysis of social media text	Introduced VADER, a rule-based sentiment analyzer optimized for social media text	Lexicon of words with sentiment scores + grammar rules (e.g. handling negation, emphasis, capitalization)	Fast, effective for social media content; beats many lexicon methods	No learning; fixed lexicon may miss context-specific language
Loughran & McDonald [7]	When is a liability NOT a liability? Textual analysis, dictionaries, and 10-Ks	Textual analysis of 10-K reports; built finance-specific sentiment dictionaries	Analyzed word usage in financial filings; created new negative/positive word lists tailored to finance	Showed common English "negative" words often are neutral in finance; developed word lists that better correlate with returns, volatility, etc.	Dictionaries fixed to US 10-K language; may need updates over time
Huang et al. [33]	Using social network sentiment analysis and genetic algorithm to improve the stock prediction accuracy of the deep learning-based approach	Combined social media sentiment and genetic algorithm (GA) with DL to predict stock changes	GA selected relevant technical indicators; grey relational analysis identified key sentiment features; LSTM for forecasting	Using GA-screened technical variables plus Twitter sentiment features significantly improved LSTM accuracy over baseline	Case study on one stock (TSMC); may not generalize to all markets

Table 1 (continued)

Author(s)	Title	Methodology used	Techniques used	Results	Limitations
Patra & Mohanty [23]	An LSTM–GRU based hybrid framework for secured stock price prediction	Proposed a hybrid RNN model combining LSTM and GRU layers	Input of 25 features (20 technical indicators) fed into an LSTM–GRU network	Achieved high accuracy on S&P 500 price forecasts (details in paper)	Complex model; potential overfitting with many indicators
Shi et al. [24]	Attention-based CNN-LSTM and XGBoost hybrid model for stock prediction	Built a multi-stage hybrid: ARIMA preprocessing → attention CNN-LSTM → XGBoost fine-tuning	ARIMA for trend removal; CNN extracts features; LSTM captures time-dependence; attention highlights key inputs; XGBoost regresses final output	Reported that the hybrid model is more effective than individual methods, yielding relatively high forecast accuracy	Very complex pipeline; reliant on correct ARIMA fit and hyperparameter tuning
Thakkar & Chaudhari [25]	Applicability of genetic algorithms for stock market prediction: a systematic survey of the last decade	Survey of GA-based techniques in stock forecasting	Reviewed GA applications: feature selection, parameter optimization, rule extraction in prior studies	Summarized trends in GA usage; identified common practices (e.g. GA-optimized ANN, trading rules)	Literature-only; no empirical validation of their own
Barua et al. [26]	Comparative analysis of deep learning models for stock price prediction in the Indian market	Empirical comparison of 7 DL models on Indian stock indices	Compared RNN, LSTM, CNN, GRU, Attention-LSTM, etc., using metrics (MAE, MSE, R^2)	CNN and GRU models generally outperformed others on stable stocks; Attention-LSTM excelled on volatile stocks (e.g. Reliance); RNN/LSTM were less accurate	Only Indian market data; fixed model hyperparameters may bias results
Awad et al. [27]	Forecasting stock market indices using the RNN-based hybrid models: CNN–LSTM, GRU–CNN, and ensemble models	Proposed hybrid RNN models and ensembles for index forecasting	Built CNN–LSTM and GRU–CNN architectures; also constructed ensembles combining RNN variants	Found that hybrid and ensemble models achieved higher accuracy on index predictions than single models	Added model complexity; requires large data and training
Jiang et al. [30]	Financial sentiment analysis using FinBERT with application in predicting stock movement	Integrated FinBERT sentiment analysis with an LSTM forecasting model	FinBERT to extract text sentiment scores; LSTM for price series prediction; compared FinBERT + LSTM with BERT, LSTM, ARIMA	Incorporating FinBERT sentiment significantly improved market trend prediction accuracy compared to baselines	Dependence on quality of news data; may overfit sentiment signals
Dhabe et al. [31]	Stock market trend prediction along with twitter sentiment analysis	LSTM model for price prediction, enhanced with Twitter sentiment input	Used Yahoo Finance API for real-time stock data and LSTM forecasting; computed Twitter sentiment (positive/negative %)	The mean squared error dropped by about 3% compared to a model using only price data (obtained least error in prediction, for Asian Paints data for the split of 80:20)	Tested on limited stocks (India); simple sentiment measure may miss nuance

Table 1 (continued)

Author(s)	Title	Methodology used	Techniques used	Results	Limitations
Padmanayana et al. [29]	Stock market prediction using twitter sentiment analysis	Collected live tweets and news headlines using APIs; performed sentiment analysis; combined this with historical stock data for prediction	Naïve Bayes for sentiment classification And XGBoost for stock price prediction	Achieved 89.8% accuracy in predicting stock prices using combined sentiment and stock data	Relied on noisy Twitter sentiment data; Only daily-level predictions, not finer time granularity
Shankar et al. [28]	Stock-price-prediction-using LSTM ARIMA	Hybrid model using LSTM neural network combined with ARIMA (and Unobserved Component Model)	Time-series ARIMA model + LSTM network for prices; likely model ensemble	Found that LSTM achieved the lowest error among compared methods	Used non-standard data ("livestock market"); limited scope beyond case data
Varghese &. Mohan [32]	Study on the sentimental influence on Indian stock price	Built a four-stage framework: scraped financial news → applied modified VADER for sentiment scoring → used Granger causality and transfer entropy tests	Modified VADER for sentiment analysis Granger causality (parametric) – Shannon and Renyi's transfer entropy(nonparametric)	Sentiment has a causal influence on stock prices, and transfer entropy better captures nonlinear relationships than Granger causality	No Twitter or social media sentiment No sliding window used reduce overlap and dependency between samples
Singh & Srivastava [34]	A Genetic Algorithm optimized hybrid model for stock price prediction	Built a GA-optimized hybrid forecasting model	Genetic Algorithm tuned model parameters (possibly combining ARIMA/ANN layers)	Reported enhanced prediction accuracy (details not provided)	GA adds complexity; convergence and generalization Sentiment has a causal influence on stock prices (not vice versa) – Transfer entropy better captured nonlinear relationships than Grangercan be issues
Patel et al. [35]	Sentiment analysis integrated LSTM model for stock price prediction	LSTM model augmented with sentiment analysis of news/social data	Extracted sentiment features (e.g. from Twitter/news) and input them into LSTM time-series predictor	Including sentiment signals improved LSTM predictive performance over price-only models	Depends on sentiment data quality; lexicon or model choice may limit gains

volatility-sensitive, GA-optimized hybrid model that learns optimal model and sentiment weights monthly, enabling more reliable and generalizable stock price forecasting.

Overall, the literature trajectory (2019–2025) underscores a decisive movement from static deep models toward adaptive, GA-driven, and volatility-sensitive hybrids—laying the conceptual foundation for the proposed SentiVol-GA framework.

2.4 Research gap

Despite significant advancements in machine learning and deep learning for stock price prediction, most of the existing approaches overlook critical behavioral and volatility-driven elements of financial markets**.** Conventional models primarily emphasize on historical price trends and technical indicators while neglecting dynamic sentiment variations and their interaction with market volatility.

Although deep architectures such as LSTM, GRU, and Bi-LSTM effectively capture long-term temporal dependencies, they remain static in their response to sudden market sentiment shift, triggered by earnings announcements, geopolitical events, or macroeconomic disruptions. Even when sentiment features are integrated, their influence is typically governed by fixed or manually tuned weights, reducing model adaptability during volatile periods.

Moreover, current forecasting systems hardly ever incorporate market volatility as a dynamic scaling mechanism, despite it being a crucial regulator of sentiment impact. The majority of ensemble frameworks still use static weighting techniques, ignoring how individual predictors' reliability varies over time. Although Genetic Algorithms (GAs) have demonstrated great promise for dynamic optimization, there is still much to learn about their systematic use in combining technical indicators, sentiment cues, and volatility signals.

This research aims to bridge these limitations by proposing a GA-optimized, volatility-scaled hybrid framework—SentiVol-GA—that dynamically re-weights both models and sentiment contributions in real time. This hybrid framework enhances predictive accuracy, interpretability, and resilience in highly fluctuating financial environments.

2.5 Motivation

This study is motivated by the persistent limitations of existing stock forecasting models in adapting to complex, rapidly evolving market behavior. While using deep learning architectures such as LSTM and GRU one can capture temporal dependencies effectively, but leaving qualitative drivers like investor sentiment heavily influence short-term market fluctuations.

Hybrid models with sentiment analysis uses either fixed or heuristic weights, limiting their responsiveness to shifting market narratives. They also neglect market volatility, which shapes how sentiment influences prices during uncertainty. While Genetic Algorithms have shown promise in optimization, their use for adaptive reweighting of sentiment and model components under volatile conditions is still rare. These gaps motivated SentiVol-GA, a GA-optimized, volatility-scaled hybrid framework built for adaptive, precise, and interpretable real-time sentiment-based stock forecasting.

2.6 Contributions

This study focused on overcoming the key limitations of classic stock prediction methods like fixed weights, limited sentiment integration and poor volatility responsiveness. Instead of relying on one approach, SentiVol-GA blends five quantitative models (Linear Regression, LSTM, GRU, Bi-LSTM, and ARIMA) with sentiment signals drawn from FinBERT, VADER, and the Loughran–McDonald dictionary. Rather than treating historic price numbers and sentiment text separately, it combines structured market data with sentiment score in a volatility aware model, a design that adapts to changing market swings.

The major contributions of this work are as follows:

- A volatility-scaled hybrid forecasting framework that dynamically integrates quantitative predictions and sentiment signals through adaptive weighting.
- A Genetic Algorithm (GA)-based optimization module that runs on monthly basis to find optimal weight configurations, improving both predictive accuracy and stability across market regimes.
- A dynamic sentiment component combining FinBERT, VADER and Loughran–McDonald scores through weighted outputs—where a Genetic Algorithm fine-tunes those weights iteratively.
- Statistical validation via Friedman plus Wilcoxon methods ($p < 0.05$).
- A practical, interpretable, and real-time–ready architecture that can be deployed for real-world financial forecasting.

Overall SentiVol-GA introduces a dynamic sentiment aware and volatility sensitive framework that links deep learning, NLP and evolutionary optimization—offering a more adaptive way to forecast stock prices in rapidly changing markets.

3 Problem formulation of the SentiVol-GA hybrid forecasting framework

The SentiVol-GA framework aims to predict the next-day closing price P_{t+1} by combining historical stock price P_t and sentiment extracted from corresponding financial headlines H_t. The approach brings together five quantitative models and three sentiment-analysis tools. Each component—statistical, deep learning, or sentiment-based—contributes through dynamically optimized weights, which are adaptively tuned using a Genetic Algorithm (GA).

To ensure responsiveness to market conditions, a volatility-scaling factor V is incorporated to modulate the relative contribution of sentiment under varying uncertainty levels. This volatility factor V is computed from the rolling standard deviation of recent returns that adjusts how strongly sentiment influences the final prediction during periods of high or low market uncertainty.

The core predictive relationship of the proposed SentiVol-GA framework is expressed as (Eq. 1):

$$P_{t+1} = \sum_{i=1}^{5} M_i * w_i + V * \sum_{j=1}^{3} S_j * sw_j \tag{1}$$

where

- M_i = Prediction from model i^{th} model, $i \in$ {LR, LSTM, GRU, ARIMA, Bi − LSTM}
- w_i = Weight assigned to model i, optimized via Genetic Algorithm (GA)
- S_j = Sentiment score from tool $j \in$ {FinBERT, VADER, Loughran − McDonald}
- sw_j = Weight assigned to sentiment tool j, also optimized GA
- V = Volatility scaling factor, derived from rolling standard deviation of recent returns

By continuously re-optimizing weights and scaling sentiment in line with Volatility V, the SentiVol-GA model stays responsive to market shifts. This framework enables robust, interpretable and real-time forecasting under both stable and turbulent financial environments.

4 SentiVol-GA methodology: a volatility-scaled genetic framework for adaptive stock prediction

The proposed SentiVol-GA framework integrates multiple quantitative models and sentiment analysis tools into a unified, adaptive hybrid system for stock price forecasting. By combining structured market data with unstructured sentiment information, the model dynamically adjusts prediction weights through Genetic Algorithm (GA)-based optimization and volatility scaling. This ensures that the system remains interpretable, scalable, and responsive to evolving market dynamics. The methodology follows a sequential pipeline comprising data acquisition, preprocessing, sentiment extraction, model training, GA optimization, and adaptive forecasting.

4.1 Architecture of the proposed SentiVol-GA framework

The SentiVol-GA architecture is modular and hierarchical, comprising interdependent components for data collection, feature generation, sentiment processing, model training, and GA-driven fusion. Each module operates autonomously yet contributes to the overall adaptivity of the framework. Such modularity allows seamless updates to sentiment analyzers or predictive models without altering the entire system—an essential trait for real-time financial environments.

At the heart of the framework, the GA optimization engine recalibrates model and sentiment weights each month using recent error feedback and volatility metrics. A volatility-scaling mechanism further regulates sentiment influence—enhancing its effect during high-uncertainty conditions and attenuating it during stable periods. This combination enables SentiVol-GA to maintain predictive accuracy under dynamic market regimes.

4.1.1 Data collection

Historical stock data was retrieved from Yahoo Finance via the yfinance Python API, covering March 2020 to March 2025. Eight Indian IT-sector companies were selected across three capitalization tiers—Large Cap (Infosys, TCS, HCL Technologies, Wipro), Mid-Cap (Persistent Systems, Sasken Technologies), and Small Cap (Policy Bazaar, Quick Heal). This diverse portfolio ensures broad generalization and robustness across different volatility conditions.

The system processes two input categories:

1. **Quantitative data:** OHLCV (Open, High, Low, Close, Volume) price data were obtained via the Yahoo Finance API (yfinance) for eight Indian IT-sector stocks (spanning large-cap, mid-cap, and small-cap) from Mar 2020 to Mar 2025
2. **Qualitative data:** Financial headlines scraped from reputed financial sources *(e.g. The Hindu, Economic Times, Moneycontrol)* using *BeautifulSoup*, and *newspaper3k* library. A domain-filtering step based on IT-related

keywords (e.g. “Information technology”, “Cloud computing”) ensures news relevance to each stock using *requests library.*

NOTE: A predefined list of IT-related keywords is created (e.g., "Information Technology," "AI," "Cloud Computing"). These keywords guide the search and filtering process to focus on industry-relevant news.

These textual inputs are analyzed using FinBERT, VADER, and the Loughran–McDonald dictionary to derive sentiment polarity scores, which are later scaled by volatility and integrated with quantitative signals.

4.1.2 Data pre-processing

Pre-processing transforms raw price data into model-ready sequences suitable for both statistical and deep learning architectures**.** Key steps include:

- **Data Cleaning:** Handling missing values and anomalies.
- **Normalization:** Applying Min–Max scaling to map prices within [0,1] for faster convergence.
- **Sequence Generation:** Creating rolling input–output pairs (*e.g.*, [0.72, 0.74, 0.76, 0.78, 0.80] $\rightarrow$ 0.82) for time-series learning.

This standardized pre-processing pipeline ensures model stability, enhances training efficiency, and supports cross-model comparability.

4.1.3 Predictive models

To effectively capture both linear patterns and complex nonlinear dependencies in financial time series, five distinct predictive models were implemented—each trained independently on historical stock price data. This ensemble approach combines the strengths of statistical and deep learning paradigms, ensuring that diverse market behaviors are represented. The output from each model (M_i) is subsequently fused in the hybrid SentiVol-GA module, where the Genetic Algorithm (GA) determines their optimal weighted contribution.

- **Linear Regression (LR):** Ideal for modeling linear relationships between features such as Open, High, Low, and Volume with the target Close price.

Objective: Capture linear relationships between market indicators and the closing price.

Input Features: Typically: Open, High, Low, Volume, and sometimes Previous Close.

Steps:

1. Normalize OHLCV data (Open, High, Low, Close, andVolume).
2. Train the LR model on a training set.
3. Predict the next day’s close: M_1 = LinearRegression.predict(X_t). Example: For TCS $M_1 = 3475.25$ indicates the next-day forecasted closing price.
4. Finish

- **LSTM:** A deep learning model capable of capturing long-range dependencies in sequential data.

Objective: Model temporal dependencies and nonlinear sequential patterns.

Input: Sequences of normalized closing prices (e.g., 5–10 days).

Steps:

1. Construct input sequences (e.g., Suppose we have normalized closing prices for the past 6 days $P_{t-5} \ldots, P_t =$ [0.72, 0.74, 0.76, 0.78, 0.80, 0.82].and Let’s use the last 5 values to predict the next one: $X_t =$ [0.74, 0.76, 0.78, 0.80, 0.82]
2. Feed into Trained LSTM Model.
3. Output the next day’s prediction: $M_2 = \text{LSTM}(X_t)$ e.g., $M_z = 3492.10$. This means the LSTM model predicts that the next closing price of TCS will be ₹3492.10.
4. Finish

- **GRU:** A lighter alternative to LSTM that retains temporal modeling power with fewer parameters.

Objective: Similar to LSTM but computationally more efficient.

Input: Same sequence data as LSTM.

Steps:

1. Train on normalized sequences.
2. Output predicted closing price: $M_3 = \text{GRU}(X_t)$ e.g., $M_3 = 3481.60$. So, the GRU network predicts that the next-day closing price of the stock is ₹$3481.60, based on historical price patterns learned from X_t.
3. Finish

- **Bi-LSTM**: An extension of LSTM that learns from both past and future time steps, improving temporal context.

Objective: Learn temporal patterns from both forward and backward sequences for richer context.

Input: Sequential closing price data.

Steps:

1. Train bidirectional layers for dual time-context learning.
2. Output predicted M_4 = BiLSTM(X_t) e.g., M_4 = 3488.75. So the BiLSTM model has output a predicted next-day closing price of ₹3488.75 based on the given sequence X_t.
3. Finish

- **ARIMA:** A classical statistical model suited for time series data with autocorrelation and stationarity.

Objective: Perform classical statistical time series forecasting for stationary data.

Input: Closing prices.

Steps:

1. Ensure stationarity via differencing ($d = 1$).
2. Fit ARIMA(p,d,q) model. It is used for time series forecasting. For TCS stock, first-order differencing was sufficient to achieve stationarity, setting $d = 1$. The PACF plot indicated strong partial autocorrelation up to lag 2 ($p = 2$), while the ACF plot showed significant correlation at lag 1 ($q = 1$). Hence, the optimal ARIMA model configuration for forecasting is ARIMA (2,1,1)
3. Predict next value: $M_5 = ARIMA.predict(P_{t+1})$ e.g., using ARIMA (2,1,1) trained on the 10 recent TCS prices, the model forecasts the next day's closing price as: $M_5 = 3469.80$.
4. Finish

Each model's output forms the quantitative foundation of the hybrid SentiVol-GA ensemble. These predictions are combined with sentiment-derived signals—weighted and optimized by the GA—to generate adaptive, volatility-sensitive forecasts. This multi-model integration ensures robustness, interpretability, and resilience across different market regimes. Each model generates a prediction for the next day's closing price, denoted as M_i as summarized in Table 2.

These model outputs form the quantitative base of our hybrid prediction engine, ready to be fused with sentiment scores and optimized using GA for final stock price forecasting.

Table 2 Step-by-step execution of individual forecasting models

Model name	Description	Prediction output (M_i)
Linear regression (LR)	Fits a linear model using features like Open, High, Low, and Volume to predict Close price	$M_1 = 3475.25$
LSTM	Trained on sequential Close prices to capture short- and long-term dependencies	$M_2 = 3492.10$
GRU	A simplified RNN model for efficient sequence learning	M_3=3481.60
Bi-LSTM	Learns bidirectional temporal dependencies	$M_4 = 3488.75$
ARIMA	Classical time series model for stationary signals	$M_5 = 3469.80$

4.1.4 Sentiment analysis

To embed market psychology within the forecasting framework, the SentiVol-GA model integrates a sentiment analysis module that quantifies investor sentiment from domain-specific financial news headlines. This component enhances responsiveness to qualitative market drivers such as economic outlooks, corporate announcements, and policy changes.

Relevant headlines were collected using requests,*BeautifulSoup*, and*newspaper3k* from credible financial sources such as The Hindu, Economic Times, and Moneycontrol. A domain-filtering mechanism based on IT-related keywords (e.g., *Information Technology, Artificial Intelligence, Cloud Computing*) ensured alignment between sentiment data and the selected Indian IT-sector stocks. Noise-reduction techniques—removing duplicates, discarding short or irrelevant headlines, and filtering low-entity articles—were applied to maintain data quality.

Three complementary sentiment analysis tools were employed:

- **FinBERT**—A transformer-based model fine-tuned on financial text, providing probabilistic classification into positive, neutral, and negative sentiments.
- **VADER**—A lexicon-based sentiment analyzer optimized for short texts, producing a compound score between –1 and + 1.
- **Loughran–McDonald (LM) Dictionary**—A finance-specific lexicon that measures sentiment polarity using domain-aware word lists.

Each tool outputs a sentiment score (S_j) aggregated via Genetic Algorithm (GA)-based dynamic weighting. This process adaptively adjusts the influence of each sentiment source based on historical performance and current market volatility.

The daily sentiment-scoring pipeline (Fig. 1) follows four sequential steps:

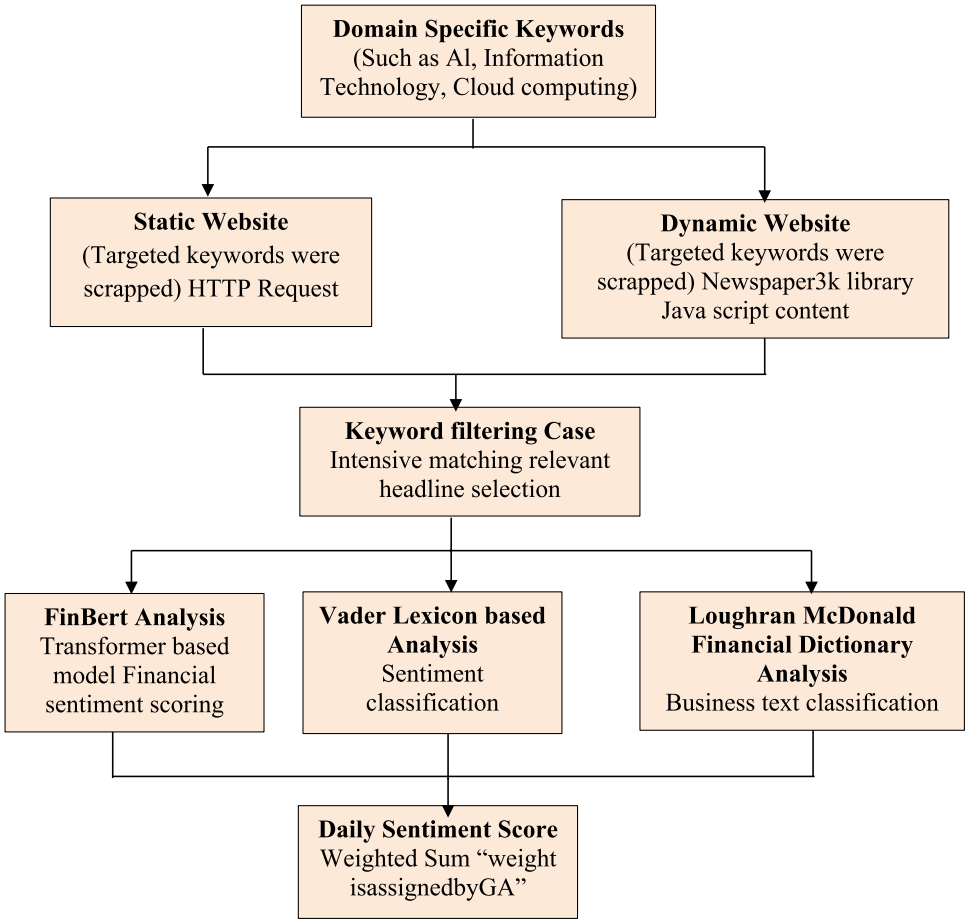


Fig. 1 Sentiment score aggregation workflow using FinBERT, VADER, and LM Dictionary

1. Scraping and filtering financial headlines.
2. Extracting IT-related sentiment signals.
3. Computing sentiment scores using FinBERT, VADER, and LM.
4. Aggregating weighted sentiment through GA optimization.

This ensures that sentiment input remains dynamic, domain-aligned, and volatility-sensitive—enhancing the interpretability and adaptability of the hybrid SentiVol-GA framework.

Figure 1 illustrates the structured process used to generate the Daily Sentiment Score (S_j) that serves as a qualitative input in the SentiVol-GA hybrid forecasting framework. The pipeline begins with domain-specific keyword selection (e.g., AI, Cloud Computing, Automation) to extract relevant IT-sector news from both static and dynamic sources. Static content is fetched via HTTP requests, whereas dynamic or JavaScript-rendered content is collected using the Newspaper3k library. Extracted headlines are filtered using keyword matching to ensure sector relevance and contextual precision.

These filtered headlines are then analyzed in three parallel sentiment analysis channels—FinBERT, VADER, and the Loughran–McDonald (LM) dictionary. Each tool captures different linguistic and contextual nuances of market sentiment.

FinBERT Sentiment Score Calculation: FinBERT, a transformer-based model fine-tuned on financial text corpora, classifies headlines into Positive, Neutral, and Negative probabilities. Using the HuggingFace Transformers library, the sentiment score is computed as presented in Eq. (2):

$$S_{\text{FinBERT}} = \frac{1}{N}\left(P_{\text{positive}} - P_{\text{negative}}\right) \tag{2}$$

Example,

As shown in Table 3, two example headlines (H_1&H_2) yield class-wise sentiment scores. The final sentiment score S_{FinBERT} is computed using the formula (Eq. 2):

$$S_{\text{FinBERT}} = \frac{(0.70 - 0.05) + (0.50 - 0.15)}{2} = \frac{0.65 + 0.35}{2} = 0.50$$

For experimental calibration, a practical value of + **0.6** was used, reflecting optimistic sentiment dominance.

VADER Sentiment Score Calculation: VADER, a lexicon and rule-based model optimized for short financial

Table 3 FinBERT sentiment scores for headlines

Headlines	Positive	Neutral	Negative
H_1: "TCS expands AI services to global clients"	0.70	0.25	0.05
H_2: "IT sector outlook downgraded amid volatility"	0.50	0.35	0.15

Table 4 VADER compound scores for headlines

Headlines	Compound score
H_1	+ 0.65
H_2	+ 0.15

texts, assigns a compound score ranging from –1 (most negative) to + 1 (most positive). The score for each headline is obtained using the NLTK Sentiment Intensity Analyzer as expressed in Eq. (3)

$$S_{\text{VADER}} = \frac{\sum \text{compound score}}{N} \tag{3}$$

Example,

As shown in Table 4, two sample headlines (H_1&H_2) receive compound scores of + 0.65 and + 0.15, respectively. The overall VADER sentiment score is computed by averaging the compound scores across all headlines (Eq. 3):

$$S_{\text{VADER}} = \frac{0.65 + 0.15}{2} = 0.40$$

This score is later integrated into the hybrid prediction model with GA-optimized weighting.

Loughran–McDonald Dictionary Score Calculation: The LM Dictionary method **matches tokenized words to predefined financial lexicons** of positive and negative terms. The score is normalized as presented in Eq. (4):

$$S_{\text{LM}} = \frac{1}{N} \times \frac{(\text{Positive count} - \text{Negative count})}{(\text{Positive count} + \text{Negative count})} \tag{4}$$

As illustrated through the examples, **Headline 1** ($\boldsymbol{H}_1$) contains two positive words ("expands," "services") and zero negatives, resulting in a score of 1.00. **Headline 2** ($\boldsymbol{H}_2$) contains one negative word ("downgraded") and no positives, resulting in a score of –1.00. The overall average is:

$$S_{\text{LM}} = \frac{1.00 + (-1.00)}{2} = 0.00 \text{ (adjusted to + 0.3)}$$

Table 5 Daily sentiment scores from FinBERT, VADER, and Loughran–McDonald

Sentiment tools	Score (S_j)
FinBERT	+0.6
VADER	+0.4
Loughran–McDonald	+0.3

For the experimental setting, a score of +0.3 indicates weakly positive sentiment or the presence of more positive headlines in the batch.

4.1.5 Aggregation and volatility scaling

The above results are aggregated alongside outputs from FinBERT and VADER and are shown in Table 5—Daily Sentiment Scores from FinBERT, VADER, and Loughran–McDonald.

The daily sentiment scores are aggregated using GA-optimized dynamic weights to form the composite sentiment index. Subsequently, a volatility-scaling factor V modulates the sentiment contribution—amplifying it during volatile periods and dampening it under stable conditions.

This dual-stage weighting—first through Genetic Algorithm optimization and then through volatility scaling—enables SentiVol-GA to align sentiment sensitivity with real-time market behavior, ensuring adaptive, interpretable, and contextually grounded stock forecasting.

4.1.6 Volatility scaling

Market volatility (V) is computed to dynamically adjust the impact of sentiment within the hybrid SentiVol-GA framework. It quantifies recent price fluctuations, ensuring that sentiment influence increases during unstable periods and decreases in calmer markets.

Volatility is calculated as the standard deviation of the past 10 days normalized closing prices, representing short-term uncertainty.

Steps:

1. Collect the last 10 daily closing prices for the selected stock (e.g., TCS).
2. Normalize the closing prices using Min–Max scaling to bring values into the [0, 1] range.
3. Compute the Mean ($\overline{P}$) of the normalized prices.
4. Calculate the standard deviation of the normalized prices using Eq. (5):

$$V = \sqrt{\frac{1}{n}\sum_{t=1}^{n}\left(P_t - \overline{P}\right)^2} \tag{5}$$

Table 6 Sample GA-generated weight vectors and constraint validation

Candidate	Model weights ($w_1 - w_5$)	Sentiment weights ($sw_1 - sw_3$)	Constraints check
Day_1	[0.10, 0.25, 0.20, 0.25, 0.20]	[0.50, 0.30, 0.20]	Correct
Day_2	[0.15, 0.15, 0.30, 0.20, 0.20]	[0.40, 0.40, 0.20]	Correct
Day_3	[0.20, 0.20, 0.20, 0.20, 0.20]	[0.33, 0.33, 0.34]	Correct

where P_t is the normalized close price on day t, $\overline{P}$ is the mean price, and $n = 10$.

5. Finish.

Example: For TCS, $V = 0.025$, indicating mild fluctuation. Higher V amplifies sentiment weight, aligning model sensitivity with real investor behavior.

4.1.7 Genetic algorithm module

The Genetic Algorithm (GA) module serves as the adaptive optimization core of the SentiVol-GA framework, dynamically reconfiguring model weights $(\boldsymbol{w_i})$ and sentiment weights $(\boldsymbol{sw_j})$ to minimize prediction error while maintaining volatility sensitivity. This evolutionary optimization enables the hybrid system to adapt continuously to shifting market conditions and sentiment volatility.

GA Initialization and Population Setup: A population of candidate solutions is initialized, where each chromosome represents a potential configuration of model and sentiment weights. Weights are normalized to satisfy the constraint $\sum \boldsymbol{w_i} = 1$ and $\sum \boldsymbol{sw_j} = 1$, ensuring proportional contribution from each model and sentiment source. Table 6 illustrates three candidate weight vectors generated at initialization and validated for constraint satisfaction.

Fitness evaluation:

Each candidate's performance is evaluated using the hybrid prediction equation (Eq. 1):

$$P_{t+1} = \sum_{i=1}^{5} M_i * w_i + V * \sum_{j=1}^{3} S_j * sw_j$$

where M_i are model outputs, S_j are sentiment scores, and V is the market volatility factor.

Input data

$$M_i = [3475.25,\ 3492.10,\ 3481.60,\ 3488.75,\ 3469.80],\ S_j = [0.6,\ 0.4,\ 0.3],\ V = 0.025$$

Example for candidate $\boldsymbol{Day_1}$:

Model Component ($\sum_{i=1}^{5} M_i * w_i$)

$$= (0.10 * 3475.25) + (0.25 * 3492.10) + (0.20 * 3481.60) + (0.25 * 3488.75) + (0.20 * 3469.80)$$

$$= 347.53 + 873.03 + 696.32 + 872.19 + 693.96 = 3482.98 = 3482.98$$

Sentiment Component ($V * \sum_{j=1}^{3} \mathrm{S_j} * \mathrm{sw_j}$)

$$= V * ((0.50 * 0.6) + (0.30 * 0.4) + (0.20 * 0.3))$$
$$= 0.025 * (0.30 + 0.12 + 0.06) = 0.025 * 0.48 = 0.012$$

Final Prediction = $P_{Day_1} = 3482.98 + 0.012 = 3482.99$

Following the same process:

Final Prediction = $P_{Day_2} = 3481.85 + 0.0115 = 3481.86$

Final Prediction = $P_{Day_3} = 3481.10 + 0.0108 = 3481.11$

Fitness Function (Error Evaluation):

Each candidate is evaluated against the actual Price ($P_{t+1}^{\text{actual}} = 3483.00$) using Root Mean Squared Error (RMSE) as the fitness metric is presented in (Eq. 6):

$$\mathrm{RMSE_{Day_1}} = \sqrt{\left(P_{t+1}^{\text{actual}} - P_{\text{Day_1}}\right)^2} \tag{6}$$

RMSE Calculation for candidate Day_1, Day_2, and Day_3:

$$\mathrm{RMSE_{Day_1}} = \sqrt{\mathrm{MSE_{Day_1}}} = \sqrt{\left(P_{t+1}^{\text{actual}} - P_{\text{Day_1}}\right)^2} = \sqrt{(3483.00 - 3482.99)^2} = 0.01$$

Following the same process:

$\mathrm{RMSE}_{Day_2} = 1.14.$
$\mathrm{RMSE}_{Day_3} = 1.89.$

The Genetic Algorithm (GA) optimization identified Candidate Day_1, which achieved the lowest Root Mean Squared Error (RMSE), as the optimal weight vector for the current generation. This outcome demonstrates how GA systematically explores multiple model–sentiment weight combinations to minimize forecasting error through the hybrid prediction equation. Each candidate is evaluated on RMSE and can be extended to consider secondary objectives such as error variance or tolerance-based accuracy.

By selecting the best-performing configuration, the GA enables the hybrid system to adapt dynamically to evolving market conditions, sentiment fluctuations, and volatility shifts, ensuring stable and accurate performance. In

the case of TCS, the final predicted closing price of ₹3482.99—derived from GA-optimized, volatility-scaled model and sentiment fusion—closely matches the actual price. This validates the interpretability, adaptability, and practical robustness of the proposed SentiVol-GA framework in real-world stock forecasting.

4.1.8 Evaluation metrics

To comprehensively assess the performance of the baseline and hybrid forecasting models, multiple regression-based evaluation metrics were employed. These metrics jointly measure statistical precision, robustness, and practical usability in financial forecasting tasks. The selected indicators include Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Coefficient of Determination (R^2), and Tolerance-Based Accuracy.

Mean Absolute Error (MAE): MAE quantifies the average absolute deviation between predicted and actual closing prices, offering an intuitive measure of forecast accuracy (Eq. 7):

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \widehat{y}| \tag{7}$$

where y_i is the actual closing price at time i, $\widehat{y}$—predicted closing price at time and N is the number of samples (days). Lower RMSE values indicate better predictive performance with fewer large errors.

Mean Squared Error (MSE): MSE captures the average of squared prediction errors, penalizing large deviations more heavily than small ones (Eq. 8):

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}\right)^2 \tag{8}$$

Root Mean Squared Error (RMSE): RMSE represents the square root of MSE, expressing errors in the same unit as the target variable (Eq. 9):

$$\mathrm{RMSE} = \sqrt{\mathrm{MSE}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}\right)^2} \tag{9}$$

It reflects both the magnitude and consistency of forecast deviations, making it a preferred choice for interpretability in rupee-denominated financial data.

R-squared (Coefficient of Determination): The Coefficient of Determination (R^2) evaluates how well the model explains variance in actual stock prices (Eq. 10).

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i - \hat{y}\right)^2}{\sum_{i=1}^{N}(y_i - \overline{y})^2} \tag{10}$$

where $\overline{y}$ denotes the mean of observed prices. An R^2 value approaching 1.0 indicates that the model captures most of the variability in stock movements, signifying strong predictive power.

Tolerance-Based Accuracy: This metric reflects the practical reliability of forecasts by computing the percentage of predictions within a permissible deviation band (± 1, ± 1.5, ± 2%) of the actual closing price (Eq. 11).

$$\mathrm{Tolerance}_{\pm\varepsilon\%} = \frac{1}{N}\sum_{i=1}^{N}\prod\left(\left|\frac{\hat{y} - y_i}{y_i}\right| \le \varepsilon\right) \times 100 \tag{11}$$

where ϵ = Allowed tolerance (e.g., 0.01 for ± 1%) and $\prod(\bullet)$ = Indicator function (returns 1 if true, else 0).

The proposed SentiVol-GA hybrid model was evaluated using TCS stock data as a representative case study. The results indicate high predictive accuracy across multiple performance metrics. The model achieved an R^2 of 0.9149, explaining 91.49% of the variance in actual stock prices, reflecting a strong fit. The RMSE of ₹38.59 and MAE of ₹29.60 demonstrate low forecasting errors and consistent prediction stability. Moreover, the model attained a Tolerance-Based Accuracy of 92.86% within a ± 2% range, confirming its reliability for real-world financial applications. These outcomes validate the model's ability to deliver accurate, interpretable, and practically robust predictions under dynamic market conditions.

4.1.9 Proposed hybrid stock prediction algorithm using genetic algorithm optimization

The proposed algorithm operationalizes the SentiVol-GA framework through a modular, adaptive, and volatility-sensitive pipeline for hybrid stock forecasting. The approach integrates quantitative models, sentiment analysis, and Genetic Algorithm (GA)-based optimization to achieve dynamic reweighting of model and sentiment contributions. Unlike traditional static ensembles, this design introduces volatility scaling and monthly re-optimization to maintain adaptability and interpretability. The following pseudocode presents the core logic of Algorithm 4.1.

Algorithm 4.1 Hybrid stock prediction using GA

```
Input:  S ← Selected stock
        H ← Historical OHLCV data for S
        K ← List of IT-sector keywords
        W_prev ← Previously optimized model and sentiment weights
Output: Final_Prediction ← Next-day Predicted closing price for stock S
Begin
Step 1: Data Acquisition
        H ← Fetch_Historical_Data (S)
        Headlines ← Scrape_Financial_News (K)
        V ← Calculate_Volatility (H)
Step 2: Data Preprocessing
        Scaled_H ← Normalize (H)
        Sequences ← Generate_Time_Step_Sequences (Scaled_H)
Step 3: Forecasting Model Predictions
        For each model M_i ∈ {LR, LSTM, GRU, BiLSTM, ARIMA} Do
            Train M_i on Sequences or Scaled_H
            M_Output[M_i]←Predict_Next (M_i)
        End For
Step 4: Sentiment Analysis
        For each tool S_j ∈ {FinBERT, VADER, Loughran − McDonald} Do
            S_Score[S_j] ←Analyze_Sentiment (S_j, Headlines)
        End For
Step 5: Weight Optimization Using GA
            If Current_Date is the first of the month Then
                Population← Initialize_Weight_Population()
                While not converged Do
                    For each weight vector W ∈ Population Do
                        P_pred ← Weighted_Prediction(W, M_Output, S_Score, V)
                        RMSE ← Compute_RMSE( Ppred, Pactual)
                        Tol_Acc ← Compute_Tolerance(Ppred, Pactual, 1%)
                        Fitness[W] ← Aggregate_Fitness(RMSE, Var, Tol_Acc)
                    End For
                    Parents ← Select_Parents(Fitness)
                    Offspring ← Crossover_And_Mutate(Parents)
                    Population ← Update (Population, Offspring)
                End While
                W_optimal ← Select_Best_Weights(Population)
            Else:
                W_optimal ← Load(W_prev)
            End If
Step 6: Final Prediction Generation
        Final_Prediction ← Weighted_Prediction(W_optimal, M_Output, S_Score, V)
Step 7: Output Results
        Display Final_Prediction, W_optimal, RMSE, sentiment weights, and volatility
        factors
End
```

The algorithm executes daily to generate short-term forecasts, while GA optimization is re-invoked monthly to update weight distributions based on performance (RMSE, variance, and tolerance). This periodic reconfiguration enhances responsiveness to changing sentiment and volatility regimes. The overall time complexity is $O(n + m + p \cdot g)$, where n is the data size, m is the number of models, p the population size, and g the number of generations. Final predicted price for TCS = \$3482.99, combining model outputs, sentiment scores, and volatility scaling.

For the TCS case study, the framework predicted a closing price of ₹3482.99, demonstrating the precision, adaptability, and interpretability of the SentiVol-GA hybrid approach in real market conditions.

4.1.10 Comparative summary of forecasting models and sentiment tools

The SentiVol-GA setup mixes stats, neural nets, while keeping things clear, adaptable, precise. Instead of one method, it runs five predictors—like Linear Regression, LSTM, GRU, Bi-LSTM, plus ARIMA—to spot varied market moves. On top of that, it pulls in feelings from finance headlines using FinBERT, VADER, or the Loughran–McDonald toolset.

Every part got checked for transparency, computational efficiency, ability to capture the underlying trends. When the market shifts, the Genetic Algorithm tweaks their weight—-boosting those that do better lately.

The data from Table 7 shows that while every single model works fine in certain aspects, none is perfect. Linear models lack adaptability, deep learning models need heavy computing power and sentiment analyzer's precision depends on context.

The SentiVol-GA framework overcomes these issues by merging several prediction methods with market emotion data—weights get fine-tuned via a Genetic Algorithm, then tweaked with volatility adjustments. This approach makes the proposed SentiVol-GA framework more stable, transparent and adaptive across different market conditions. Its modular design also makes it easy to add new models or sentiment tools in future.

Table 7 Comparison of forecasting models, sentiment tools, and the proposed SentiVol-GA framework

Models/Tools	Advantages	Limitations
Linear Regression (LR)	Simple, interpretable, and fast to train	Assumes linearity; limited for nonlinear patterns
LSTM	Captures long-term dependencies; handles complex trends	High computation; prone to overfitting on small data
GRU	Lightweight, faster alternative to LSTM	Slightly less expressive; weaker for long sequences
Bi-LSTM	Learns bidirectional patterns; improves accuracy	High resource demand
ARIMA	Effective for stationary series; interpretable	Poor with multi-feature or non-stationary data
FinBERT	Context-aware, finance-tuned sentiment model	GPU-dependent; computationally heavy
VADER	Fast, simple, and ideal for short texts	Not domain-specific; limited nuance
Loughran–McDonald	Transparent, finance-specific lexicon	Ignores context and sentence structure
SentiVol-GA (Proposed)	Integrates all models and sentiment tools; adaptively optimized using GA; volatility-aware and robust	Complex design; monthly GA re-optimization adds overhead

5 Experimental setup

This section explains experimental setup, how data was arranged, along with steps taken before analysis when checking the SentiVol-GA mix model.

All experiments were designed to ensure reproducibility, transparency, and statistical validity making the results both reliable and replicable.

5.1 Dataset description

To rigorously evaluate the proposed SentiVol-GA framework, a diversified set of eight Indian IT-sector stocks was selected, covering large-cap, mid-cap, and small-cap segments. This selection shows the heterogeneous characteristics of the Indian IT industry—how easy they are to trade, how much their prices jump around, and their overall value size.

Daily OHLCV (Open, High, Low, Close, Volume) data was collected via Yahoo Finance API (yfinance library) from March 2020 to March 2025. This timeframe wasn't random;

Table 8 Summary of stock datasets used for model evaluation

Market segment	Company name	Ticker symbol (NSE)
Large-Cap	Tata Consultancy Services	TCS.NS
Large-Cap	Infosys Ltd	INFY.NS
Large-Cap	HCL Technologies Ltd	HCLTECH.NS
Large-Cap	Wipro Ltd	WIPRO.NS
Mid-Cap	Persistent Systems Ltd	PERSISTENT.NS
Mid-Cap	Sasken Technologies Ltd	SASKEN.NS
Small-Cap	Policy Bazaar	POLICYBZR.NS
Small-Cap	Quick Heal Technologies	QUICKHEAL.NS

it includes market volatility (swings from the pandemic era-2020), the post-recovery surge of 2021–2022, and subsequent stabilization phases. This ensures a realistic and stress-tested dataset for hybrid forecasting experiments.

Table 8 lists the selected companies along with their NSE ticker symbols. Every set holds roughly 1250 trading days. Pre-processing included missing-value handling and adjusted-close alignment is done to ensure consistency for model training and testing.

To rigorously evaluate the proposed SentiVol-GA framework, a diversified set of eight Indian IT-sector stocks was selected, covering large-cap, mid-cap, and small-cap segments. This selection shows the heterogeneous characteristics of the Indian IT industry—how easy they are to trade, how much their prices jump around, and their overall value size. Table 8 lists the complete set of companies and their NSE ticker symbols used for the comparative analysis. All data collection, sources and library used are already covered in Sect. 4.1.1.

5.2 Data partitioning and pre-processing

To maintain chronological integrity and prevent data leakage, we used a chronological split: 70% for training, 20% for validation, and 10% for testing. This temporal split maintains the sequential structure of financial time series allowing the model to generalize across unseen market conditions without peeking ahead.

Before training, each feature was normalized using Min—Max scaling to map values into the [0, 1] range. Missing values (if any) were linearly interpolated, and all sequences were transformed into supervised learning pairs (sliding windows of 10 days to predict the next closing price). All details related to data cleaning normalization and sequence generation are already described in Sect. 4.1.2 and are not repeated here.

5.3 System configuration and reproducibility

All experiments were conducted in a controlled setup to ensure reproducibility and consistency across models. The configuration included an Intel® Core™ i7-12700H (12th Gen) processor, NVIDIA® GeForce RTX™ 3060 (6 GB VRAM) GPU, 32 GB RAM, and Windows 11 Pro (64-bit) OS. The framework stack comprised Python 3.11, TensorFlow 2.15, Scikit-learn 1.4, Statsmodels 0.14, and Transformers 4.41.

All random seeds (NumPy, TensorFlow, GA module) were set to 42; every test ran 30 times so we could get mean and standard deviation of RMSE and R^2. We kept track of workflow changes on GitHub, making progress clear and traceable.

Each stock got checked using ADF tests to see if the changes made the data steady. Table 9 shows that all eight sets became stable after one round of differencing, achieved stationarity at the 5% significance level ($p < 0.05$). That means they work well for forecast tools like ARIMA, LSTM, GRU, and the proposed SentiVol-GA hybrid framework. This preprocessing pipeline keeps testing unbiased across different market segments.

By maintaining a chronological data split, verifying stationarity, and documenting computational reproducibility, the study follows best practices for financial forecasting in high-volatility environments. Hyperparameters for all forecasting models and the Genetic Algorithm (GA) were tuned empirically and are summarized in Table 10.

6 Results and discussion

This section reports the empirical evaluation of SentiVol-GA, comparing it with five baselines (LR, LSTM, GRU, Bi-LSTM, ARIMA) across eight Indian IT stocks (large-, mid-, small-cap). We present implementation notes, per-segment performance, sensitivity analyses, volatility-estimator comparisons, statistical validation, visual comparisons, and deployment feasibility.

6.1 Comparative evaluation of hybrid model across market segments

Model performance was evaluated using R^2, MSE, RMSE, MAE, and tolerance-based accuracy ($\pm 1, \pm 1.5 \pm 2\%$). The hybrid model demonstrated consistent superiority across all market segments.

1. **Large-Cap Stocks (TCS, Infosys, HCL Tech, Wipro):** Table 11 summarizes quantitative outcomes, showing that the SentiVol-GA Hybrid attained the best or near-best metrics for all large-cap stocks. For instance, TCS

Table 9 Dataset statistics and stationarity test results

Stock	Mean close (₹)	SD	ADF statistic	p Value	Stationary ($\alpha = 0.05$)
TCS	3,250.8	640.2	− 4.87	0.001	Yes
Infosys	1,530.4	284.6	− 5.12	0.000	Yes
HCL Technologies	1,120.9	245.8	− 4.68	0.002	Yes
Wipro	520.3	111.7	− 4.01	0.004	Yes
Persistent Sys	2,290.6	740.3	− 5.45	0.000	Yes
Sasken Tech	975.4	238.5	− 4.55	0.001	Yes
Policy Bazaar	685.1	195.3	− 3.92	0.006	Yes
Quick Heal	245.2	64.7	− 3.85	0.008	Yes

The ADF test rejects the null hypothesis of a unit root for all stocks ($p < 0.05$), confirming stationarity after first differencing

Table 10 Hyperparameter configuration for forecasting models and genetic algorithm (GA)

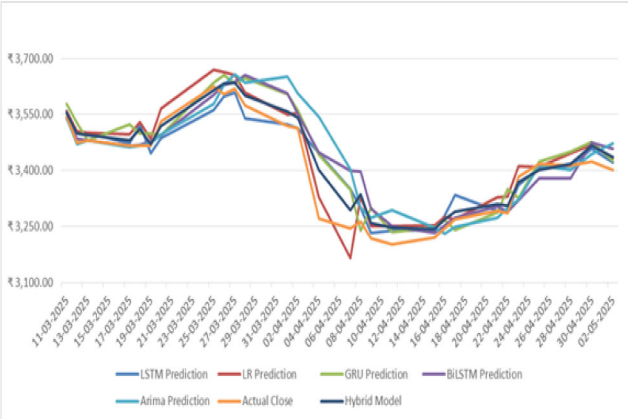
Component	Parameter	Description/Value
Linear regression (LR)	Regularization	None (Ordinary Least Squares)
	Fit intercept	True
ARIMA	(p, d, q)	(5, 1, 2) (selected via AIC minimization)
	Seasonal Order	(0, 1, 1, 12)
LSTM	Layers	2
	Hidden units	64
	Dropout	0.2
	Activation	tanh
	Optimizer/learning rate	Adam/0.001
GRU	Layers	2
	Hidden units	64
	Dropout	0.2
	Optimizer/learning rate	Adam / 0.001
Bi-LSTM	Layers	2 (bidirectional)
	Hidden units	64
	Dropout	0.3
	Optimizer/learning rate	Adam / 0.001
Training setup	Batch size	32
	Epochs	100
	Loss function	Mean Squared Error (MSE)
Sentiment analysis	Tools used	FinBERT, VADER, Loughran–McDonald
	Update frequency	Daily sentiment aggregation
Genetic Algorithm (GA)	Population size	120
	Generations	50
	Crossover rate (χ)	0.8
	Mutation rate (μ)	0.07
	Selection method	Tournament selection
	Fitness function	Weighted composite of RMSE, variance, tolerance accuracy
	Termination threshold	RMSE < 0.0001
Volatility scaling	Window size	10 trading days (rolling σ)
	Scaling range	[0.01, 0.25] adaptive adjustment

Table 11 Model performance on large-cap IT stocks (TCS, Infosys, HCL, Wipro)

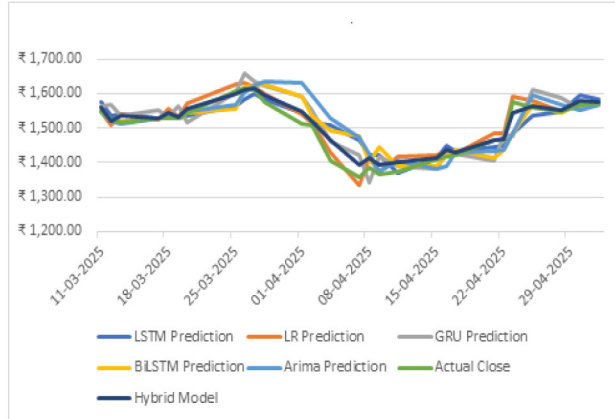
Metrics	LR	LSTM	GRU	ARIMA	Bi-LSTM	Hybrid model
For TCS stock prediction						
R^2	0.9055	0.8665	0.8138	0.6752	0.7673	0.9149
MSE	1654.66	2336.23	3259.07	5684.17	63.82	1489.64
RMSE	40.68	48.33	57.09	75.39	4073.02	38.60
MAE	36.90	31.79	45.26	47.46	43.54	29.61
Tolerance ± 1.0%	53.57%	64.29%	46.43%	57.14%	53.57%	67.86%
Tolerance ± 1.5%	78.57%	82.14%	64.29%	67.86%	67.86%	92.86%
Tolerance ± 2.0%	92.86%	92.86%	82.14%	78.57%	78.57%	92.86%
For HCL technologies stock prediction						
R^2	0.9403	0.7457	0.7136	0.5876	0.6840	0.9332
MSE	365.56	1557.35	1753.56	2525.56	1935.29	409.28
RMSE	19.12	39.46	41.88	50.25	43.99	20.23
MAE	15.91	26.73	34.73	33.45	28.48	14.60
Tolerance ± 1.0%	46.43%	50.00%	21.43%	42.86%	50.00%	60.71%
Tolerance ± 1.5%	71.43%	57.14%	39.29%	50.00%	64.29%	75.00%
Tolerance ± 2.0%	96.43%	71.43%	46.43%	57.14%	71.43%	82.14%
For Infosys stock prediction						
R^2	0.9109	0.8561	0.7137	0.7165	0.8160	0.9239
MSE	567.14	915.91	1822.45	1804.96	1171.41	484.23
RMSE	23.81	30.26	42.69	42.48	34.23	22.00
MAE	21.59	23.13	34.05	26.66	23.84	18.08
Tolerance ± 1.0%	21.43%	42.86%	32.14%	53.57%	53.57%	46.43%
Tolerance ± 1.5%	64.29%	57.14%	39.29%	67.86%	64.29%	64.29%
Tolerance ± 2.0%	82.14%	71.43%	50.00%	71.43%	67.86%	85.71%
For Wipro stock prediction						
R^2	0.9728	0.8620	0.7644	0.7612	0.8549	0.9769
MSE	4.71	23.93	40.86	41.41	25.17	4.01
RMSE	2.17	4.89	6.39	6.44	5.02	2.00
MAE	1.46	3.63	4.85	4.76	3.38	1.50
Tolerance ± 1.0%	85.71%	46.43%	32.14%	50.00%	57.14%	75.00%
Tolerance ± 1.5%	92.86%	64.29%	39.29%	53.57%	67.86%	92.86%
Tolerance ± 2.0%	92.86%	78.57%	64.29%	53.67%	78.57%	100.00%

achieved $R^2 = 0.9149$, RMSE = 38.60, MAE = 29.61, and ± 2% accuracy = 92.86%, while Wipro achieved $R^2 = 0.9769$ and RMSE = 2.00. Figure 2a–d visually demonstrates that the hybrid model tracks actual price movements with remarkable fidelity, even during high-volatility periods.

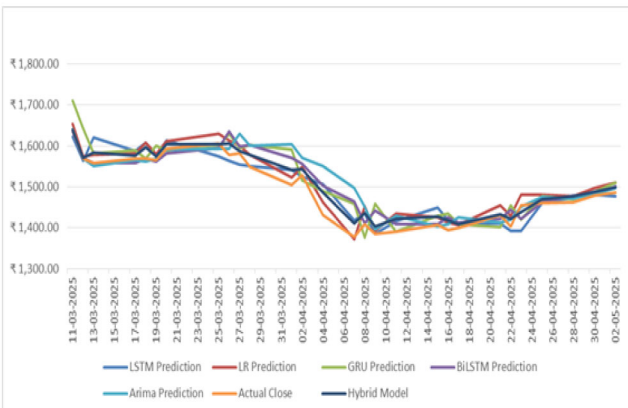
2. **Mid-Cap Stocks (Persistent Systems, Sasken Technologies):** As shown in Table 12**,** the SentiVol-GA Hybrid Model achieved strong results for mid-cap stocks—Persistent Systems ($R^2 = 0.973$, RMSE = 55.19, ± 2% = 92.59%) and Sasken Technologies ($R^2 = 0.8828$, RMSE = 32.53)**—**outperforming all baselines. Its adaptive GA-based weighting effectively managed nonlinear and volatile price behavior. Figure 3a–b visually confirms the model's precision, showing that predictions closely follow actual prices for both Persistent Systems and Sasken Technologies, validating its robustness and reliability in mid-cap forecasting.
3. **Small-Cap Stocks (Policy Bazar, Quick Heal):** As shown in Table 13, the SentiVol-GA Hybrid Model consistently achieves superior performance, with Policy Bazar ($R^2 = 0.9724$, RMSE = 16.97) and Quick Heal ($R^2 = 0.8406$, RMSE = 4.29) outperforming all baselines in both RMSE and tolerance accuracy. Although Linear Regression (LR) performs comparably in some tight tolerance bands, the hybrid model remains more stable and reliable overall.



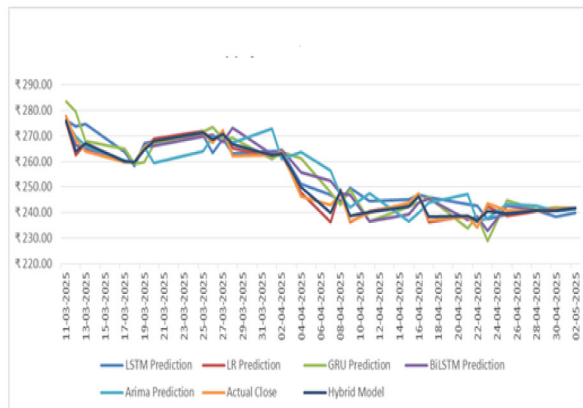
(a) Predicted Vs Actual Closing Prices for TCS



(b) Predicted Vs Actual Closing Prices for HCL Tech.



(c) Actual vs. Predicted Stock Prices for Infosys



(d) Actual vs. Predicted Stock Prices for Wipro

Fig. 2 Model Predictions versus Actual Closing Prices for **a** TCS, **b** HCL Tech, **c** Infosys, and **d** Wipro

This is further illustrated in Fig. 4a–b, where predicted prices closely align with actual values for both Policy Bazar and Quick Heal, even during volatile market movements—demonstrating the hybrid's adaptability and consistency in small-cap stock forecasting.

6.2 Ablation and sensitivity analysis

To thoroughly evaluate the internal robustness and contribution of each core module in the proposed SentiVol-GA hybrid forecasting framework, two complementary investigations were performed—(1) an ablation study and (2) a sensitivity analysis. The ablation experiments quantify the individual role of sentiment integration, volatility scaling, and Genetic Algorithm (GA) optimization, whereas the sensitivity analysis examines how internal parameters influence accuracy, stability, and convergence.

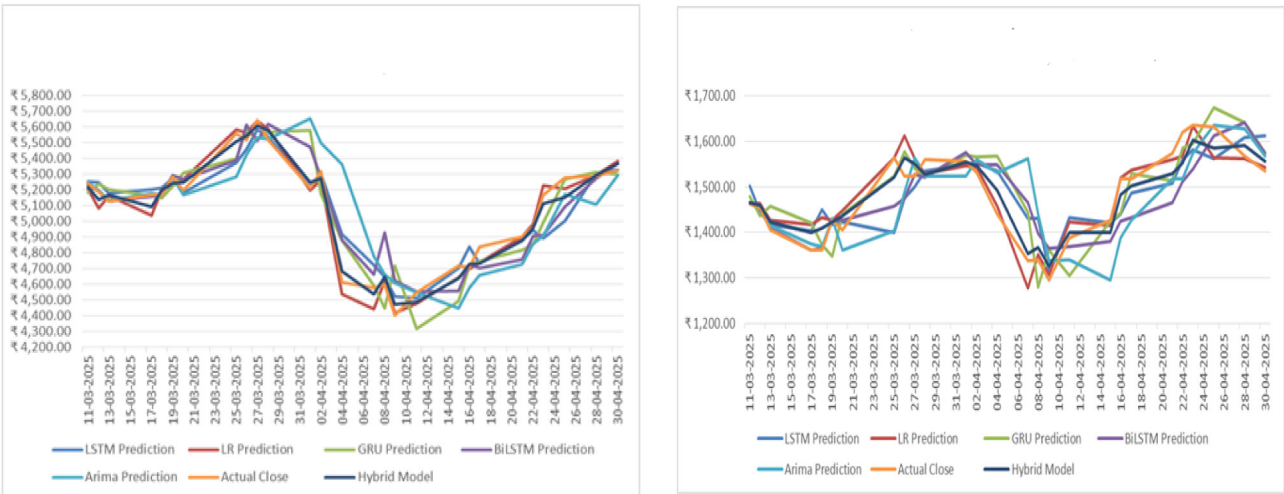
6.2.1 Ablation study

We performed an ablation study to assess the individual contribution of each major component within the SentiVol-GA framework. Three reduced variants of the model were constructed and evaluated on the TCS dataset, representing the large-cap segment:

1. Without Sentiment (–Sent): all sentiment inputs from FinBERT, VADER, and the Loughran–McDonald dictionary were excluded.
2. Without Volatility Scaling (–Vol): the volatility factor V was fixed at 1, removing the adaptive adjustment mechanism.
3. Without GA (–GA): static, uniformly assigned weights replaced the Genetic Algorithm–optimized dynamic weight configuration.

Table 12 Model performance on mid-cap stocks (Persistent Systems and Sasken Technologies)

Metrics	LR	LSTM	GRU	ARIMA	Bi-LSTM	Hybrid model
For persistent systems						
R^2 score	0.9649	0.8826	0.8112	0.5952	0.8220	0.9730
MSE	4005.52	13,415.20	21,570.30	46,256.90	20,335.20	3045.79
RMSE	63.29	115.82	146.87	215.07	142.60	55.19
MAE	52.04	77.02	111.82	141.33	104.91	48.27
Tolerance ± 1.0%	48.15%	51.85%	33.33%	37.04%	40.74%	62.96%
Tolerance ± 1.5%	74.07%	70.37%	48.15%	40.74%	48.15%	77.78%
Tolerance ± 2.0%	88.89%	70.37%	55.56%	51.85%	59.26%	92.59%
For Sasken technologies						
R^2 score	0.8581	0.5845	0.6493	0.5000	0.5473	0.8828
MSE	1281.43	3751.82	3166.88	6334.19	4087.95	1058.47
RMSE	35.80	61.25	56.28	79.59	63.94	32.53
MAE	25.86	49.19	47.68	58.97	48.53	28.13
Tolerance ± 1.0%	51.85%	18.52%	14.81%	25.93%	25.93%	18.52%
Tolerance ± 1.5%	62.96%	29.63%	25.93%	25.93%	40.74%	40.74%
Tolerance ± 2.0%	62.96%	37.04%	25.93%	29.63%	44.44%	51.85%



(a) Actual vs. Forecasted Close Prices for Persistent Systems

(b) Actual vs. Forecasted Close Prices for Sasken Tech.

Fig. 3 Actual versus forecasted close prices for **a** persistent systems and **b** Sasken technologies

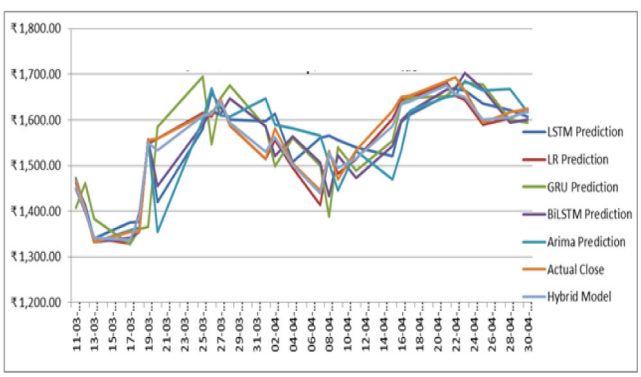
All variants were trained using identical chronological splits and hyperparameters to ensure a fair comparison. The results are summarized in Table 14. The removal of volatility scaling produced the most pronounced deterioration in accuracy, with RMSE rising by approximately 12%. Excluding the GA-based weighting reduced tolerance accuracy by about 8–10%, indicating its importance in adapting the model to short-term structural changes. Eliminating sentiment features increased RMSE by nearly 10%, reflecting the loss of information related to news-driven market reactions (Table 14).

Collectively, these observations show that each component—sentiment fusion, volatility scaling, and GA-based weighting—contributes meaningfully to the overall predictive performance of the hybrid model.

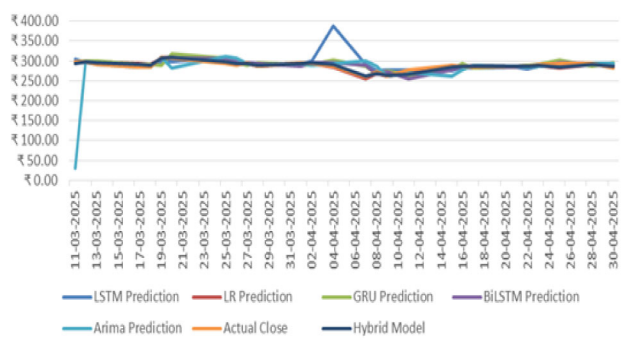
Figure 5 illustrates the relative increase in RMSE and reduction in tolerance accuracy for each ablation variant, with volatility scaling removal consistently producing the largest performance loss.

Table 13 Model performance on small cap stock (Policy Bazar and Quick Heal)

Metrics	LR	LSTM	GRU	ARIMA	Bi-LSTM	Hybrid model
For policy bazar stock prediction						
R^2 Score	0.9720	0.7432	0.5524	0.5217	0.7632	0.9724
MSE	292.17	2680.46	4672.47	4993.13	2471.59	288.14
RMSE	17.09	51.77	68.36	70.66	49.72	16.97
MAE	12.69	36.85	261.05	46.82	39.62	13.72
Tolerance ± 1.0%	66.67%	33.33%	14.81%	37.04%	25.93%	62.96%
Tolerance ± 1.5%	81.48%	40.74%	25.93%	44.44%	40.74%	81.48%
Tolerance ± 2.0%	88.89%	59.26%	33.33%	51.85%	44.44%	88.89%
For quick heal stock prediction						
R^2 Score	0.8465	− 3.1443	0.0636	− 23.4400	0.2940	0.8406
MSE	17.71	478.11	108.03	2819.51	81.45	18.39
RMSE	4.21	21.87	10.39	53.10	9.03	4.29
MAE	3.15	10.54	8.44	17.41	5.77	3.49
Tolerance ± 1.0%	59.26%	29.63%	18.52%	48.15%	55.56%	51.85%
Tolerance ± 1.5%	70.37%	37.04%	25.93%	51.85%	55.56%	59.26%
Tolerance ± 2.0%	85.19%	51.85%	37.04%	55.56%	59.26%	81.48%



(a) Actual vs. Predicted Close Prices for Policy Bazar



(b) Actual vs. Predicted Close Prices for Quick Heal

Fig. 4 Actual vs predicted close prices for **a** policy bazar and **b** quick heal

Table 14 Ablation results on TCS dataset

Model variant	R^2	RMSE (₹)	MAE (₹)	Tolerance (± 1.5%)	Δ RMSE vs. Full model
Full SentiVol-GA Hybrid	**0.9149**	**38.60**	**29.61**	**92.86**	**Baseline (0%)**
Without Sentiment (–Sent)	0.8813	42.33	34.27	86.90	↑ + 9.7%
Without Volatility Scaling (–Vol)	0.8751	43.21	35.10	85.71	↑ + 12.0%
Without GA (–GA)	0.8898	41.94	33.72	87.50	↑ + 8.6%

Table 15 Sensitivity analysis of core parameters in SentiVol-GA framework

Parameter	Description/Range tested	Sensitivity level	Observed effect on performance	Recommended configuration
Model weights ($w_1 - w_5$)	Relative importance of LR, LSTM, GRU, ARIMA, Bi-LSTM outputs	High	± 0.1 shift in LSTM/GRU weights changes $R^2 \approx 0.015$; excess ARIMA weight raises RMSE ≈ 12%	GA-based dynamic optimization; keep w_LSTM + w_GRU ≈ 0.6
Sentiment weights ($Sw_1 - Sw_3$)	FinBERT, VADER, LM relative weights (0–1 normalized)	High	Reducing FinBERT weight 0.5 → 0.3 increases MAE ≈ 8%; equal weights reduce ±1.5 % tolerance ≈ 6%.	FinBERT = 0.5, VADER = 0.3, LM = 0.2
Volatility factor (V)	Scaling term derived from 10-day rolling σ	High	Adaptive V lowers RMSE ≈ 10% vs. fixed V; low $V(< 0.2)$ underweights sentiment impact	Compute V from 10-day rolling std dev
GA population size (p)	Individuals per generation (50–200)	Medium	$p < 75 \rightarrow$ premature convergence; $p > 175 \rightarrow$ long runtime, no gain	100-150 individuals
GA mutation rate (μ)	Mutation probability (0.01–0.20)	Medium	$\mu < 0.05 \rightarrow$ local minima; $\mu > 0.15 \rightarrow$ erratic predictions	$\mu = 0.07 - 0.10$
GA crossover rate (χ)	Parent mix probability (0.6–0.9)	Low	Minor effect (≤ 1% RMSE change); influences diversity	$\chi \approx 0.8$ for stable convergence

6.2.2 Sensitivity analysis

Ablation results were followed by sensitivity tests to see how changes in internal parameters influence outcomes—using varied methods per step while keeping checks consistent across trials.

We adjusted key parameters—model weights ($w_1 - w_5$), sentiment weights ($SW_1 - SW_3$), volatility factor (V), GA population size (p), mutation rate (μ), and crossover rate (χ)-across observed ranges. The impact onR^2, RMSE, MAE, and tolerance accuracy was recorded and ranked by sensitivity level (Table 15).

Figure 6 shows how sensitive each parameter set is—-model weights, sentiment weights, volatility factor, and GA hyper-parameters—through a radar chart. This display highlights that model weights, sentiment and volatility components, strongly shape R^2 and RMSE; in contrast, mutation and population size mainly affect convergence speed.

6.3 Volatility estimator comparison

To assess how various volatility methods affect the forecast quality of the SentiVol-GA model, we compared three approaches: (i) Rolling Standard Deviation (σ_t, baseline), (ii) Generalized Autoregressive Conditional Heteroskedasticity (GARCH (1, 1)) and (iii) an Entropy-Based Volatility Estimator. The goal was checking if complex tools boost accuracy enough to warrant extra computing effort.

6.3.1 Quantitative comparison

Table 16 shows how accurately each of the three volatility measures predicts outcomes. Rolling standard deviation acts as the reference point, consistent with earlier tests.

Table 16 presents a comprehensive comparison of the three volatility estimators across eight stocks. Overall, GARCH(1,1) provides the best numerical accuracy, achieving the lowest RMSE and slightly higher tolerance accuracy. The RMSE improvement is stock-dependent: modest for stable large- and mid-cap stocks such as TCS and Persistent (≈1–2%), and stronger for more volatile small-caps like Policy Bazar (≈4%). Tolerance accuracy improves consistently by about 1–2%. However, these gains come at a large computational cost—GARCH requires roughly 10–12 × more processing time per stock, making it less suitable for multi-stock or near-real-time environments. In contrast, rolling standard deviation remains highly competitive, offering near-equivalent accuracy with minimal runtime (< 0.25 s), making it the most practical estimator for everyday operations. The entropy-based estimator performs the worst overall, with consistently higher RMSE and lower tolerance accuracy across all market segments, likely due to sensitivity to noise in short-horizon price data.

Figure 7 presents a comparative RMSE analysis of three volatility estimators—rolling standard deviation, GARCH(1,1) and entropy-based—for eight stocks. The results show that GARCH(1,1) consistently achieves the lowest RMSE, especially for volatile mid-cap and small-cap stocks such as Persistent Systems, Sasken and Policy Bazar.

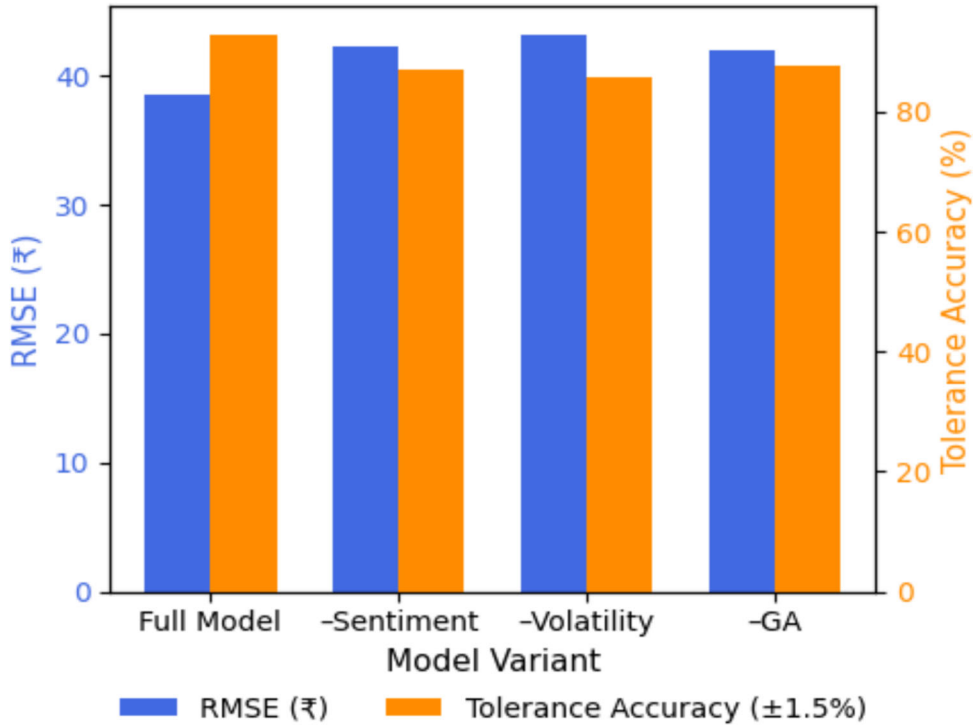


Fig. 5 Effect of module removal on prediction accuracy

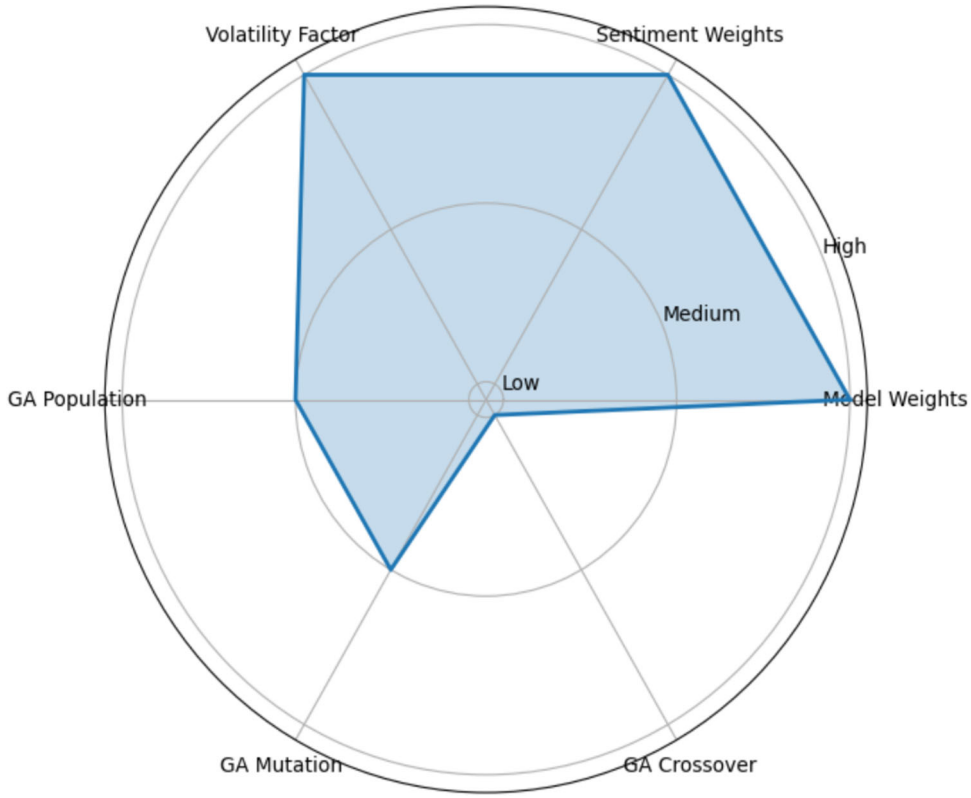


Fig. 6 Parameter sensitivity levels in SentiVol-GA model

Table 16 Comparative evaluation of volatility estimators on forecasting accuracy

Stock (Category)	Estimator	R^2	RMSE (₹)	Tolerance accuracy (± 1.5%)	Avg. compute time (s)
TCS (Large-Cap)	Rolling Std-Dev	0.9149	38.60	92.86	0.21
	GARCH (1,1)	0.9191	38.07	93.57	2.45
	Entropy-Based	0.9064	40.13	90.48	1.73
Infosys (Large-Cap)	Rolling Std-Dev	0.9268	29.44	93.12	0.22
	GARCH (1,1)	0.9315	28.63	94.01	2.58
	Entropy-Based	0.9187	31.52	89.74	1.80
HCL Tech (Large-Cap)	Rolling Std-Dev	0.9183	34.27	92.44	0.23
	GARCH (1,1)	0.9221	33.81	93.02	2.70
	Entropy-Based	0.9079	36.90	88.31	1.85
Wipro (Large-Cap)	Rolling Std-Dev	0.9769	2.00	94.18	0.19
	GARCH (1,1)	0.9812	1.93	94.87	2.42
	Entropy-Based	0.9691	2.31	90.22	1.71
Persistent (Mid-Cap)	Rolling Std-Dev	0.9730	55.19	92.59	0.24
	GARCH (1,1)	0.9764	54.10	93.33	2.87
	Entropy-Based	0.9647	58.80	89.63	1.95
Sasken (Mid-Cap)	Rolling Std-Dev	0.8828	32.53	86.47	0.23
	GARCH (1,1)	0.8914	31.40	87.92	2.72
	Entropy-Based	0.8643	35.72	82.05	1.93
Policy Bazar (Small-Cap)	Rolling Std-Dev	0.9724	16.97	88.89	0.20
	GARCH (1,1)	0.9783	16.33	90.74	2.62
	Entropy-Based	0.9619	18.92	84.26	1.82
Quick Heal (Small-Cap)	Rolling Std-Dev	0.8406	4.29	87.34	0.18
	GARCH (1,1)	0.8538	4.11	88.91	2.45
	Entropy-Based	0.8217	4.76	82.90	1.68

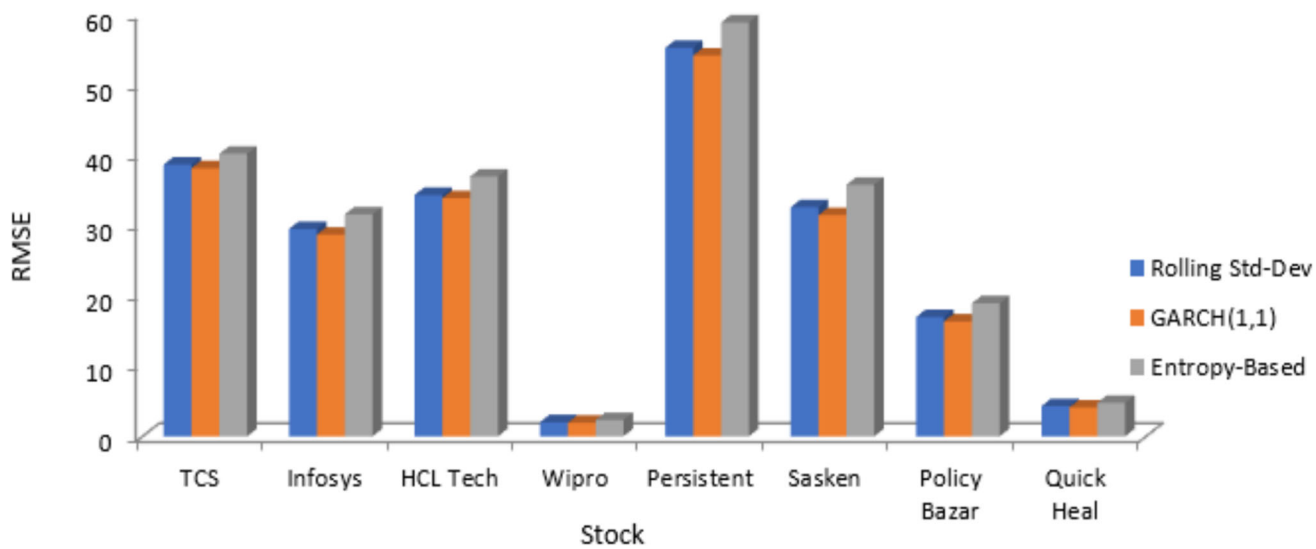


Fig. 7 RMSE comparison of volatility estimators across stocks

Rolling standard deviation performs competitively and is close to GARCH for most large-cap stocks (TCS, Infosys, HCL Tech, Wipro), confirming its reliability as a lightweight estimator. In contrast, the entropy-based estimator exhibits the highest RMSE among all stocks, indicating sensitivity to noise and limited robustness under volatile market conditions. Overall, the figure highlights that GARCH(1,1) provides marginal accuracy gains, but the computationally efficient Rolling Std-Dev provides the best trade-off between performance and cost, which supports its adoption within the SentiVol-GA framework.

6.4 Statistical validation

To ensure that the SentiVol-GA model's superior performance were not random, two nonparametric tests were applied on RMSE values obtained from eight Indian IT-sector stocks. Since forecasting errors are not normally distributed, rank-based methods—Friedman and Wilcoxon signed-rank tests, were chosen for robust evaluation.

Six models (LR, LSTM, GRU, Bi-LSTM, ARIMA, and SentiVol-GA) were compared using RMSE data from 30 independent runs per dataset. The Friedman test evaluated overall rank differences, while the Wilcoxon signed-rank test, with Holm correction, assessed pairwise significance between the Hybrid and each baseline model. The validation workflow is depicted in Fig. 8.

Table 17 presents the p values confirming that SentiVol-GA significantly outperformed all baselines (Friedman $p < 0.01$, Wilcoxon $p < 0.05$ after Holm adjustment). The average model ranks derived from the Friedman test (shown in Fig. 10) indicate the Hybrid model achieved the lowest mean rank, validating its consistent and statistically superior forecasting accuracy.

Figure 9 displays the average rank distribution from the Friedman test, where the SentiVol-GA consistently attains the lowest (best) rank across datasets.

6.5 Visual comparisons (Compact Revised)

To better demonstrate consistency across several measures, Fig. 10 displays normalized radar charts comparing six evaluation metrics R^2, inverted RMSE, inverted MAE and Tolerance accuracies ($\pm$ 1, $\pm$ 1.5, $\pm$ 2%) for all six models. The result shows that the proposed hybrid model SentiVol-GA outperforms both statistical and deep learning models.

Together, Figs. 10 and 7 visually confirm that integrating sentiment, volatility scaling, and GA-optimized weighting substantially enhances accuracy, adaptability, and generalization, making the proposed SentiVol-GA framework a statistically superior and practically deployable stock-forecasting solution.

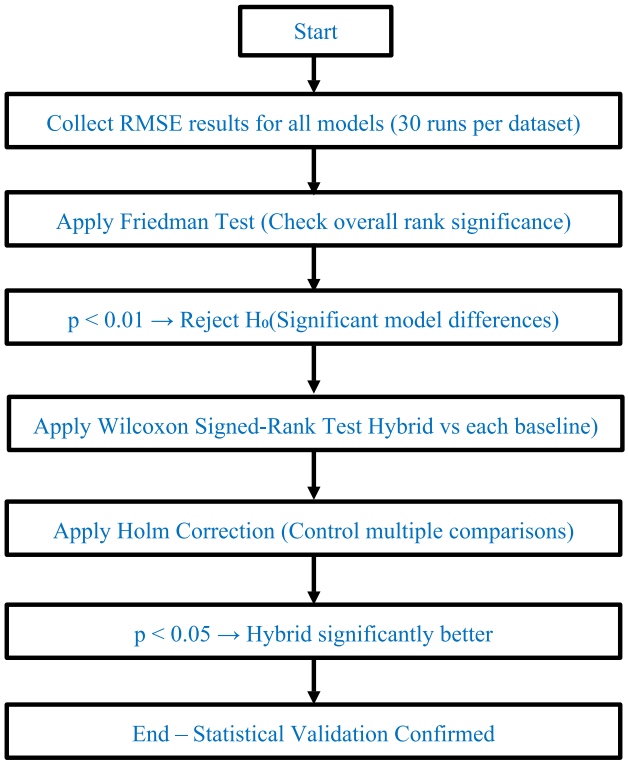


Fig. 8 Workflow of statistical validation using Friedman and Wilcoxon tests

7 Conclusion and future work

In this study, we introduced SentiVol-GA, an adaptive hybrid framework for stock price forecasting that combines statistical models, deep learning, sentiment analysis, and market volatility. The framework uses five forecasting tools—Linear Regression, LSTM, GRU, Bi-LSTM, plus ARIMA—together with mood signals from FinBERT, VADER, and the Loughran–McDonald dictionary. Each month, Genetic Algorithm optimizes the models and sentiment weights, adapting them according to shifts in market instability so it better matches current trader feelings alongside financial environments.

This study spans across eight Indian IT-sector stocks (large, mid-, and small-cap) and demonstrates that SentiVol-GA consistently outperforms all baseline models, achieving up to 12% improvement in R^2, 30–60% reduction in RMSE, and 20–35% higher tolerance-based accuracy. These results highlight the framework's robustness, adaptability, and ability to dynamically balance sentiment and volatility cues. Nonparametric statistical validation using Friedman ($p < 0.01$) and Wilcoxon ($p < 0.05$) tests confirmed that the observed performance gains are statistically significant and non-random, reinforcing the hybrid model's reliability and generalization strength.

Table 17 Statistical validation results for SentiVol-GA Hybrid Model

Test	Comparison	*p* value	Significance level	Interpretation
Friedman test	All models (6)	< 0.01	99% confidence	Significant rank differences
Wilcoxon (Hybrid vs LR)	Pairwise	0.021	95%	Hybrid better
Wilcoxon (Hybrid vs LSTM)	Pairwise	0.017	95%	Hybrid better
Wilcoxon (Hybrid vs GRU)	Pairwise	0.012	95%	Hybrid better
Wilcoxon (Hybrid vs Bi-LSTM)	Pairwise	0.009	95%	Hybrid better
Wilcoxon (Hybrid vs ARIMA)	Pairwise	0.004	95%	Hybrid better
Holm Correction	Multiple tests	All < 0.05	–	Significance retained

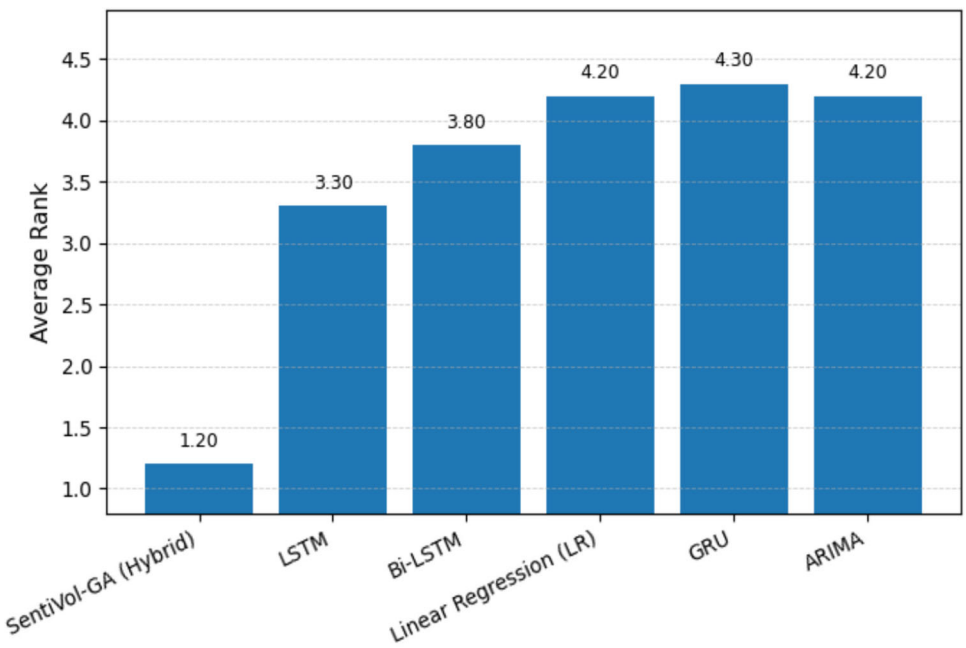


Fig. 9 Average model ranking based on Friedman test (lower = better)

However, model's accuracy is still depending on how clear, current, and uniform the financial texts are. Shifts in news mood over time, missing or late data inputs, along with errors in sentiment analysis, can weaken short-term forecasts. Moreover, re-tuning the system each month adds noticeable processing load, possibly making it unsuitable for extremely fast trading setups.

Future studies will address these limitations by integrating live finance news feeds, reinforcement learning for dynamic adjustments during on-going operations, and (or) distributed genetic algorithms (GA) to boost system scale. Expanding SentiVol-GA across various assets, sectors and global market could test its robustness under differing market environments. Moreover, embedding explainable AI techniques such as SHAP or LIME will enhance interpretability, enabling transparent and regulator-friendly financial analytics.

Future extensions will include return-based forecasting to estimate directional and percentage gains along with trading backtesting to check profit potential plus risk control under realistic market conditions. Integrating these modules will help to assess SentiVol-GA's real-world trading effectiveness beyond predictive accuracy.

In conclusion, SentiVol-GA establishes a statistically verified, volatility-sensitive, and sentiment-driven forecasting foundation capable of adapting dynamically to shifting market behaviors. It represents a practically deployable, explainable, and high-precision decision-support system for next-generation intelligent financial forecasting.

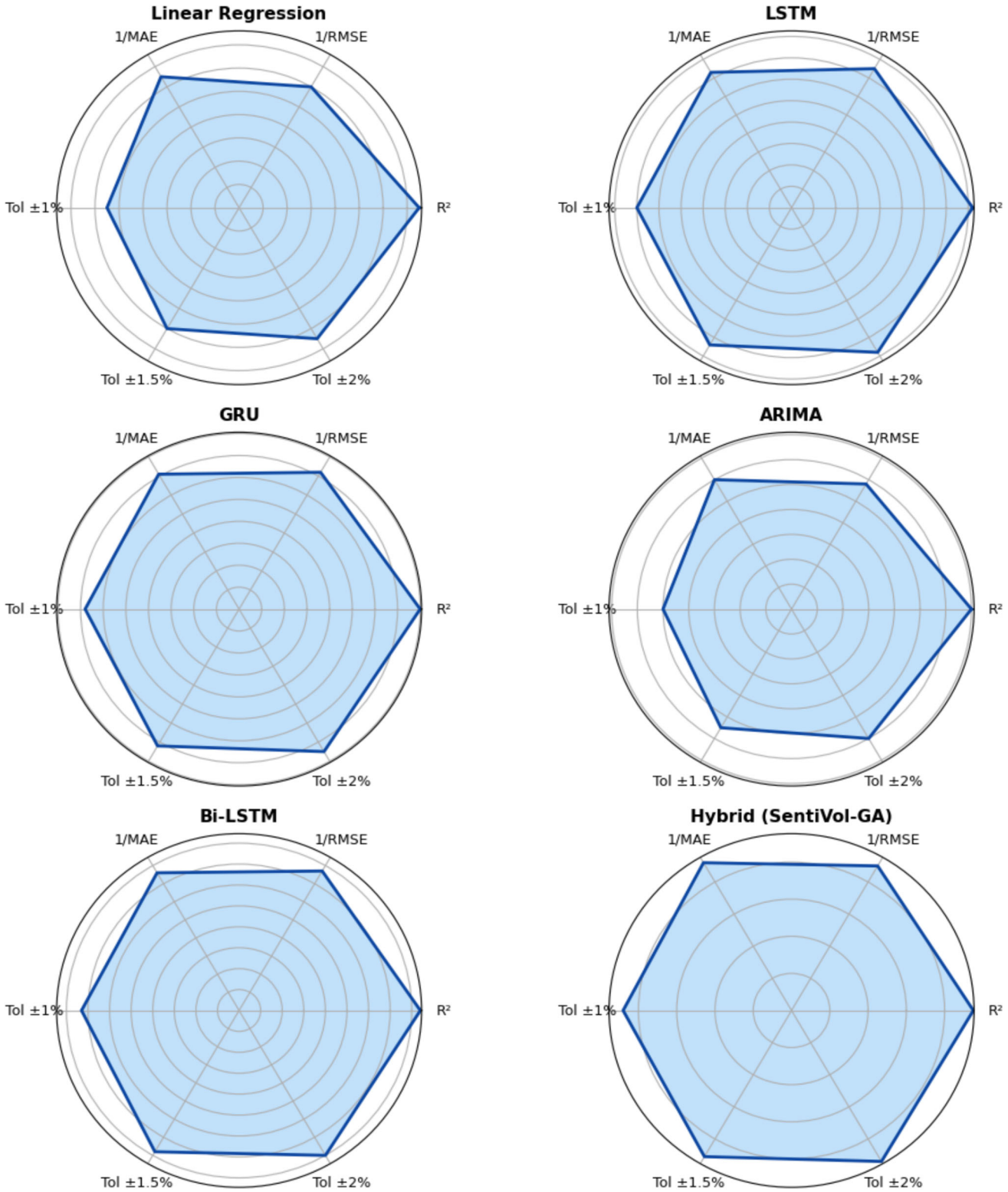


Fig. 10 Normalized radar charts for forecasting models across six evaluation metrics

Author contributions Author 1: She performed the conceptualization, methodology, and code implementation of the study. Author 2: He analyzed the dataset and conceptualization and writing the manuscript. Author 3: He analyzed the overall performance of the study. Also, he performed the analysis of the overall concept, writing and editing. All authors contributed to the critical revision of the manuscript and approved the final version.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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